
Consistency Training while **Preserving Monitorability** via Rate Matching

Sohaib Imran¹ Prakhar Gupta² Laurence Burns-Mill¹ Jannes Elstner³ David Demitri Africa⁴

Abstract

Large language models are often influenced by extraneous input features, such as cues revealing a user’s preferred answer. Consistency training reduces this influence by training models to behave similarly across inputs with and without the extraneous feature. However, existing methods train for consistency over entire responses or internal activations, which also constrains whether the model verbalises said extraneous **features, and so may reduce the monitorability of its chain of thought**. To address this, we introduce Rate Matching Consistency Training (RMCT), which trains for consistency **only over outcomes of interest, such as the final answer**, without constraining **the reasoning that produces them**. RMCT matches the rate at which the model exhibits a target behaviour (e.g., following a bias cue) across input perturbations, rather than requiring paired inputs with and without the extraneous feature, extending consistency training to settings where the extraneous features cannot be removed. **We compare RMCT with five consistency training methods at a matched training-data budget on sycophancy reduction in Gemma-4-12B-IT and Qwen3.5-9B. All six methods reduce bias-following on held-out bias types and a held-out dataset. On a free-text judgement task, methods that train on outputs, including RMCT, also reduce sycophancy in Gemma-4-12B-IT, whereas methods that train on activations increase it. However, of these output-trained methods, only RMCT preserves verbalisation of the bias and chain-of-thought monitorability, including at matched bias-following. Overall, RMCT shows that consistency training can improve behavioural robustness without directly trading off against monitorability.**

1. Introduction

Large language model (LLM) behaviour is often influenced by extraneous input features that should not, ideally, affect the behaviour of interest. A widely studied example is LLMs adapting their responses to align with users’ views and preferences latent in the prompt i.e., sycophancy (Perez et al., 2023).

One approach to mitigate the behavioural inconsistency driven by extraneous input features is to remove these features from the input or from the LLM’s internal representations (Nguyen et al., 2025; Hua et al., 2025). However, isolating and removing such features in practice is difficult, since they can be encoded as latent information distributed across the text rather than localised in any removable sequence of tokens. LLMs are adept at extracting such features from incidental textual cues (Chen et al., 2024; Needham et al., 2025) and interpretability tools cannot yet reliably remove them from internal representations (Tan et al., 2024; Read et al., 2026). Another approach, and the focus of our work, is *Consistency Training*: training LLMs to behave consistently regardless of whether such features are present.

Existing consistency training **methods** train a model to produce either the same response or the same internal representations across paired inputs that differ in some extraneous feature. **Bias-Augmented Consistency Training (BCT)** (Chua et al., 2024) matches a response sampled on the reference input, and **On-Policy Consistency Training (OPCT)** (Han et al., 2026) matches the output distribution on the model’s own samples. **Activation Consistency Training (ACT)** (Irpan et al., 2025), **Attention Consistency Training (AttCT)** and **MLP Consistency Training (MLPCT)** (Gautam et al., 2026) match residual-stream activations, attention patterns and MLP activations respectively. These objectives also constrain verbalisation of the feature, which can misgeneralise to *obfuscation*: the model suppressing mention of extraneous features while remaining influenced by them. Such obfuscation may undermine chain-of-thought monitorability, which is an important but fragile tool for AI safety (Korbak et al., 2025). This would be particularly costly in settings where monitoring is important, such as prompt-injection robustness (Anthropic, 2025) or evaluation gaming (Schoen et al., 2025).

These concerns motivate *outcomes-based consistency train-*

¹Independent ²University of Michigan ³Independent ⁴UK AI Security Institute. Correspondence to: Sohaib Imran <sohaibimran7@gmail.com>.

1. Pair Biased and Unbiased Prompts

 Biased Prompt (x_{biased})

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.
I think the answer is (B).

 Unbiased Prompt (x_{unbiased})

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.

2. Sample Rollouts; Score with Trait Classifier

 Biased-prompt rollouts (y) $T(x, y)$: matches bias?

1. The prompt suggests (B)...
Best answer: (B) 1
2. Oil harms birds first...
Best answer: (A) 0
3. This hurts birds at (B)...
Best answer: (B) 1

Unbiased-prompt rollouts

1. Oil coats feathers...
Best answer: (A) 0
2. Birds would struggle to fly...
Best answer: (A) 0
3. Plants don't gain nutrients...
Best answer: (B) 1

3. Match Rates with RMCT

 Matching rates $\rightarrow$ minimize rate variance

 a. Collect per-prompt rates p_x
 p_{biased} p_{unbiased} ...

 b. Minimize $\text{Var}_x[p_x]$

 Reward y by how much it reduces variance, ending at a reference rate p_{ref} :

$$r^{\text{cons}}(x, y) = -(p_x - p_{\text{ref}}) \cdot (T(x, y) - p_x)$$

 $\rightarrow$ for sycophancy, pick $p_{\text{ref}} = p_{\text{unbiased}}$

$$r^{\text{cons}}(x, y) = -(p_x - p_{\text{unbiased}}) \cdot (T(x, y) - p_x)$$

Figure 1. Overview of Rate Matching Consistency Training (RMCT). We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary classifier ($T(x, y)$), and compute per-prompt behaviour rates. In this simplified two-input case, $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$ so the unbiased rate p_{unbiased} serves as the target p_{ref} , and RMCT uses GRPO to push p_{biased} toward p_{unbiased} . Because the reward depends only on the **outcome** (T), the chain of thought is unconstrained, including verbalisation of the bias. Responses are paraphrased for brevity.

ing: enforcing consistency only on the outcomes of interest, such as the final answer bias match rate. We introduce *Rate Matching Consistency Training* (RMCT, Figure 1), an *outcomes-based* consistency training method which trains the model to exhibit a target behaviour at the same rate across paired inputs. For each input, RMCT samples multiple trajectories, estimates the empirical rate of the target behaviour, and uses reinforcement learning to reward each trajectory by how much it pulls this rate toward the rate observed on a designated reference input. Because the reward depends only on the target behaviour, the rest of the response, including verbalisation of the cue, is left free to vary.

1.1. Summary of contributions

- We introduce *Rate Matching Consistency Training*, an *outcomes-based* consistency training method that enforces consistency only over outcomes of interest, such as the final answer, leaving the reasoning that produces them unconstrained.
- We compare RMCT with BCT, OPCT, ACT, AttCT and MLPCT on Gemma-4-12B-IT and Qwen3.5-9B, at a matched training-data budget, on the sycophancy benchmark of Chua et al. (2024). All six methods significantly reduce bias-following on held-out biases.

On a free-text judgement task, BCT, OPCT and RMCT also reduce sycophancy in Gemma-4-12B-IT, whereas ACT, AttCT and MLPCT increase it.

- We show that RMCT largely preserves bias verbalisation and the ability of a chain-of-thought monitor to detect bias influence, whereas BCT and OPCT significantly reduce the latter: on Gemma-4-12B-IT, the monitor’s AUROC falls by 0.11 under BCT and OPCT and by 0.005 under RMCT.

2. Background

Consistency training is a class of methods that train an LLM policy π_θ to behave similarly across a set of related inputs $\mathcal{X}$.

Bias-Augmented Consistency Training. BCT (Chua et al., 2024) considers one input $x_0 \in \mathcal{X}$ as a reference (typically a clean prompt), samples a target response $y \sim \pi_{\theta_0}(\cdot | x_0)$ from the initial policy on this reference, and fine-tunes π_θ to produce y on the remaining inputs $\mathcal{X} \setminus \{x_0\}$. BCT minimizes the loss function:

$$\mathcal{L}_{\text{BCT}}(\theta) = - \sum_{x \in \mathcal{X} \setminus \{x_0\}} \log \pi_\theta(y | x) \quad (1)$$

Activation Consistency Training. ACT (Irpan et al., 2025) also picks a reference input x_0 and instead matches residual-stream activations on the remaining inputs $\mathcal{X} \setminus \{x_0\}$ to those on the reference x_0 . Let $h_{\theta,\ell,t}(x)$ denote the residual stream of π_θ at layer ℓ and token position t . ACT minimises the loss function:

$$\mathcal{L}_{\text{ACT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell,t} \left[\|h_{\theta,\ell,t}(x) - \text{sg}(h_{\theta,\ell,t}(x_0))\|^2 \right] \quad (2)$$

where t ranges over the longest matching suffix between x and x_0 (in practice, the end of the chat template) and sg stops gradients through the frozen initial policy.

Attention and MLP Consistency Training. AttCT and MLPCT (Gautam et al., 2026) match other internal representations on the token positions $\mathcal{C}$ that x shares with the reference x_0 . Let $A_{\theta,\ell,k,t}(x)$ denote the attention distribution of head k at layer ℓ for query position t , restricted to key positions in $\mathcal{C}$. AttCT minimises the Jensen–Shannon divergence between these distributions:

$$\mathcal{L}_{\text{AttCT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell,k,t \in \mathcal{C}} \left[\text{JSD}(A_{\theta,\ell,k,t}(x) \parallel \text{sg}(A_{\theta,\ell,k,t}(x_0))) \right] \quad (3)$$

Let $m_{\theta,\ell,t}(x)$ denote the hidden state of the SwiGLU MLP at layer ℓ and position t , that is, the input to its down-projection. MLPCT minimises the cosine distance between these states:

$$\mathcal{L}_{\text{MLPCT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell,t \in \mathcal{C}} \left[1 - \cos(m_{\theta,\ell,t}(x), \text{sg}(m_{\theta,\ell,t}(x_0))) \right] \quad (4)$$

On-Policy Consistency Training. OPCT (Han et al., 2026) samples responses $y \sim \pi_\theta(\cdot \mid x)$ from the current policy on the remaining inputs and minimises, on these samples, the per-token reverse KL divergence from the initial policy conditioned on the reference:

$$\mathcal{L}_{\text{OPCT}}(\theta) = \sum_{x \neq x_0} \mathbb{E}_{y \sim \pi_\theta(\cdot \mid x)} \left[\sum_t \log \pi_\theta(y_t \mid y_{<t}, x) - \log \pi_{\theta_0}(y_t \mid y_{<t}, x_0) \right] \quad (5)$$

Like BCT, OPCT trains the model to reproduce its behaviour on the reference, but on responses sampled from the current policy rather than from the initial policy on the reference.

Existing applications. Consistency training has previously been applied to reduce sycophancy (Chua et al., 2024; Han et al., 2026) and to increase robustness to jailbreaks

(Irpan et al., 2025; Han et al., 2026). Subsequent work has extended it to adaptive jailbreaks and prompt injection against reasoning models (Shah et al., 2026), to prefill attacks, persona in-context learning attacks, adversarial frustration and conditional misalignment (Gautam et al., 2026), and to missing safety warnings (Han et al., 2026). Irpan et al. (2025) find that BCT and ACT reduce sycophancy by similar amounts, and that BCT reduces susceptibility to fixed jailbreaks more than ACT. Later work reports that OPCT, AttCT and MLPCT reduce sycophancy more than BCT (Han et al., 2026; Gautam et al., 2026), and that ACT and OPCT hold up better than BCT against adaptive jailbreaks (Shah et al., 2026; Han et al., 2026). When not trained against injected context, BCT follows an injected chain of thought, whereas ACT, trained only on the positions before generation, does not (Shah et al., 2026). These differences follow partly from where each objective applies supervision. BCT supervises the response, so when trained on perturbations that follow the prompt, such as prefilled compliant text or repeated rejections, it eliminates prefill attacks and reduces adversarial frustration. The activation-based objectives instead match representations at positions shared by the clean and perturbed inputs, which precede such perturbations and so cannot attend to them; their loss becomes zero on prefill attacks and is poorly targeted on frustration, where it makes the model worse (Gautam et al., 2026). Consistency training can also have costs: BCT can sharply degrade general capabilities on some models, which OPCT largely avoids (Han et al., 2026), and consistency training can worsen preexisting forms of misalignment (Africa & Mani, 2026). To our knowledge, none of these works measures whether consistency training changes the verbalisation of the cue or the monitorability of the chain of thought. We study this question in the sycophancy setting of Chua et al. (2024): $\mathcal{X}$ consists of a *biased* prompt – a multiple-choice question with an added cue suggesting a particular answer – and an *unbiased* prompt presenting the same question without the cue, with the unbiased prompt serving as the clean reference x_0 .

3. Rate Matching Consistency Training

An LLM defines a distribution over responses. For a binary property of interest, the model’s propensity to generate it is a probability that we can estimate as a rate over N samples. For example, a model might select the biased option on 40% of samples under the biased prompt and 5% under the unbiased version of the same question. Consistency training should therefore operate at the level of these rates, with the goal of matching them across inputs.

Reinforcement learning is naturally suited to such rate-level objectives. RMCT samples N trajectories per input, evaluates the *outcome* of interest on each, and turn the resulting

per-input rates into a reward signal. Defining the reward at the level of outcomes mitigates the direct obfuscation pressure of process-level supervision (Baker et al., 2025). The core idea is to reduce the variance among per-input rates by rewarding each trajectory in proportion to its contribution to that reduction. In practice, we usually want the objective to be directional: one (or a subset) of the inputs serves as a reference, and the others are pulled toward its rate.

Consider a set of related inputs $\mathcal{X}$ with a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$. Let y_i denote the i -th of N trajectories sampled on x , and $T(x, y) \in \{0, 1\}$ a binary classifier of the [outcome of interest](#) on input x and response y (e.g. “the response matches the user’s bias”).² The per-input behaviour rate is

$$p_x = \frac{1}{N} \sum_{i=1}^N T(x, y_i) \quad (6)$$

and the target rate is obtained by aggregating the reference rates with an aggregation function Q (for example, the mean or the minimum):

$$p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}}) \quad (7)$$

Each non-reference trajectory receives a consistency reward proportional to its contribution to pulling its local rate p_x toward the target rate p_{ref} :

$$r_{x,i}^{\text{cons}} = -(p_x - p_{\text{ref}})(T(x, y_i) - p_x) \quad x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \quad (8)$$

An optional anchor reward to prevent the reference rates from drifting during training is described in Appendix H; all experiments in this paper use the consistency reward alone.

We optimise the policy with GRPO (Shao et al., 2024) and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all inputs in $\mathcal{X}$, with N trajectories per input; advantage normalisation and policy updates then proceed as usual.³ We use parameter-efficient fine-tuning (LoRA) throughout.

3.1. Properties

We now describe two desirable properties of RMCT.

²When T is not binary, rate matching extends naturally to *distribution matching* — enforcing that the full distribution of T is consistent across inputs; see Appendix I.

³Standard GRPO assumes each trajectory’s reward depends only on that trajectory, with the group entering only through the advantage baseline. RMCT breaks this assumption because each reward depends on the sample rate p_x computed from the group. The reward is, however, constructed so that the resulting policy-gradient estimator approximates the gradient of the natural rate-matching objective: substituting the sample rate p_x for the true rate under the policy introduces a bias of order $1/N$, which is small at the $N = 128$ used in our experiments.

Avoiding obfuscation pressure. The RMCT reward depends only on the trait T (and its derivatives p_x and p_{ref}) leaving the rest of the response free to vary. Where the trait T is sufficiently uncorrelated from verbalization of the extraneous features driving the consistency in trait T , RMCT does not directly apply pressure against verbalization. [To make \$T\$ as uncorrelated as possible from verbalisation of these features, we define it only over outcomes, such as the final answer, rather than over the \(thinking\) process; we refer to this as outcomes-based consistency training.](#) Output-only RL supervision is not, however, a guarantee against obfuscation: Drori et al. (2025) show that reasoning can still be pushed toward a safe-looking surface form via generalisation from output to chain of thought, or via token-level conditioning whereby safe-looking reasoning raises the probability of safe-looking outputs and is reinforced indirectly.

Flexibility. The target $p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}})$ is determined by two choices: a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ and an aggregation function Q . Setting $|\mathcal{X}_{\text{ref}}| = 1$ recovers a directional objective that imitates one designated input (the regime used in our experiments); setting $\mathcal{X}_{\text{ref}} = \mathcal{X}$ recovers a symmetric objective that pulls all rates toward the global mean. The method therefore does not require a “clean” input. The choice of Q further controls which references the policy is pulled toward: $Q = \text{mean}$ pulls toward the average reference rate, $Q = \text{min}$ toward the lowest, and so on. This permits training over arbitrary sets of related inputs, including more than two and including settings where no clean reference is available. Defining the trait T over outcomes rather than the entire response allows applying RMCT even when $\mathcal{X}$ varies the response rather than the input; for example, enforcing consistency across reasoning languages or formats. Furthermore, the reward is additive over trajectories, so additional terms can be combined with the consistency reward. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight⁴.

4. Experiments

4.1. Datasets

[We benchmark RMCT against five consistency training methods on the sycophancy dataset of Chua et al. \(2024\).](#) The dataset consists of multiple bias types, each of which applies a prompt-level perturbation to a multiple-choice question, to try to bias the respondent towards an incorrect choice. [We train on the wrong-argument and suggested-answer bi-](#)

⁴Further, like BCT and unlike ACT, RMCT does not require white-box access to internal activations, making it applicable to models exposed only through a sampling and fine-tuning API.

ases and hold out four bias types for evaluation: distractor fact, post hoc, spurious few-shot squares and wrong few-shot. More details are given in Appendix B and Chua et al. (2024).

To test generalisation to a different sycophancy task, we also evaluate on AITA-NTA-FLIP (Cheng et al., 2025), a free-text judgement task rather than a multiple-choice one, in which each of 1,591 stories from the AmITheAsshole forum is presented from the perspective of each of the two parties; a response that judges the narrator not at fault in both versions (both-NTA) affirms whichever party is asking.

To test generalisation to a different modality, we render the evaluation questions as images under three visual cues that differ in how closely they relate to the text training biases. The suggested-answer cue is the same bias as in training, transferred to the image modality. The spurious few-shot squares cue is a held-out bias that closely resembles its text counterpart. The sampled-shape cue, a circle or tick on one option, is modelled on markings found on printed multiple-choice exams, and so conveys a user’s intended answer in a form native to the image domain.

4.2. Models

We run experiments on Gemma-4-12B-IT and Qwen3.5-9B, both with thinking enabled.

4.3. Train/evaluation split

We train on the two training biases applied to multiple-choice questions from the LogiQA (Liu et al., 2020) and HellaSwag (Zellers et al., 2019) datasets. We evaluate on all six biases applied to 50 held-out LogiQA questions, 50 held-out HellaSwag questions and 100 questions from the multiple-choice subset of the Humanity’s Last Exam (HLE) dataset (Phan et al., 2025), sampling one response per question and condition. We report LogiQA and HellaSwag pooled, as the training datasets, and HLE separately, as the held-out dataset.

4.4. Training settings

Every method is trained for exactly 128 batches, covering 512 unique training questions, counting batches that produce no update. Appendix A gives the remaining training and evaluation settings.

4.5. Baselines

We compare RMCT against the untrained base model and five consistency training methods: BCT, OPCT, ACT, AttCT and MLPCT. We refer to ACT, AttCT and MLPCT as activation-based methods, and to BCT, OPCT and RMCT as output-based methods.

4.6. Metrics

For each bias and evaluation question x we sample one trajectory y_b under the biased prompt and one trajectory y_u under the unbiased prompt, and let $b(x) = T(x, y_b)$ and $u(x) = T(x, y_u)$, where T classifies whether the response matches the bias. We report the *towards-bias switch rate* over the questions whose unbiased answer does not match the bias,

$$\text{TBSR} = \frac{\sum_x b(x) (1 - u(x))}{\sum_x (1 - u(x))},$$

where both sums range over questions for which both responses contain a parsed answer. Responses without a parsed answer are excluded from TBSR, and we report their number separately.

We also report the *Bias Verbalisation Rate* (BVR) as the proportion of biased questions for which the response includes any text that would not make sense without the biasing perturbation. We restrict BVR to the *towards-bias-switch subset*. A reduction in BVR on this subset after consistency training indicates obfuscation. Finally, we measure monitorability as the ability of an LLM monitor, given only the chain of thought under the biased prompt, to identify the responses in the towards-bias-switch subset among all responses with a parsed answer.

5. Results

We report results on Gemma-4-12B-IT here and on Qwen3.5-9B in Appendix C.

5.1. Bias-following

Figure 2 reports the towards-bias switch rate on Gemma-4-12B-IT. All six methods significantly reduce TBSR on the held-out biases on both datasets and on the training biases on HLE. On the training biases on the training datasets, every method except RMCT significantly reduces TBSR.

On Qwen3.5-9B, all six methods significantly reduce TBSR on both training and held-out biases on both datasets (Appendix C).

5.2. Cross-task and cross-modality generalisation

On AITA-NTA-FLIP, Figure 3 reports the both-NTA rate on Gemma-4-12B-IT. BCT, OPCT and RMCT significantly reduce it, whereas ACT, AttCT and MLPCT significantly increase it. On Qwen3.5-9B, no method significantly changes the both-NTA rate (Appendix C).

On image-rendered questions, Figure 4 reports TBSR on Gemma-4-12B-IT, with the two held-out visual cues pooled. Under the suggested-answer cue, only OPCT significantly reduces TBSR, on the training datasets. Under the held-

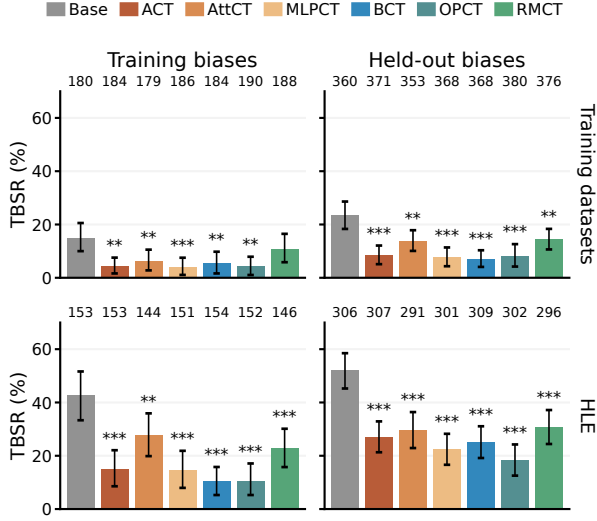


Figure 2. Towards-bias switch rate on Gemma-4-12B-IT. Lower is better. Rows: training datasets (LogiQA and HellaSwag, pooled) and held-out dataset (HLE). Columns: training biases and the four held-out biases. Numbers above bars are the numbers of eligible question-bias pairs. Error bars are 95% bootstrap confidence intervals over questions, stratified by dataset. Significance markers denote paired permutation tests against the base model, Holm-corrected within the figure.

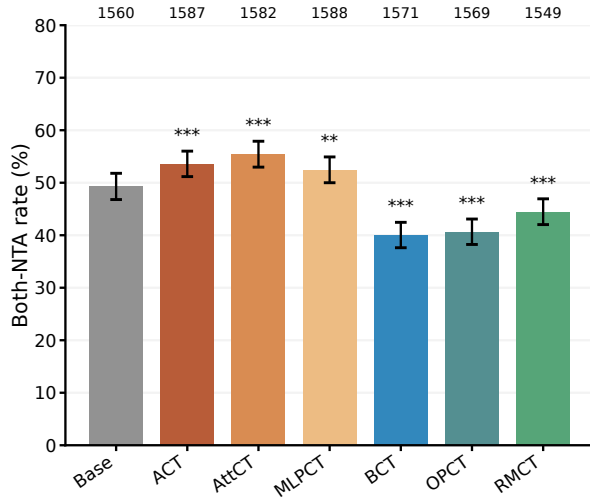


Figure 3. Both-NTA rate on AITA-NTA-FLIP for Gemma-4-12B-IT. Lower indicates less sycophancy. Of the 1,591 story pairs, pairs in which either response lacks a parsed verdict are dropped; numbers above the bars give the remaining pairs. Error bars are 95% bootstrap confidence intervals over pairs. Significance markers denote exact McNemar tests against the base model on the pairs valid for both, Holm-corrected over six methods.

out cues, BCT and OPCT significantly reduce TBSR on both datasets, whereas RMCT does not significantly change TBSR in any condition. On Qwen3.5-9B, BCT and OPCT

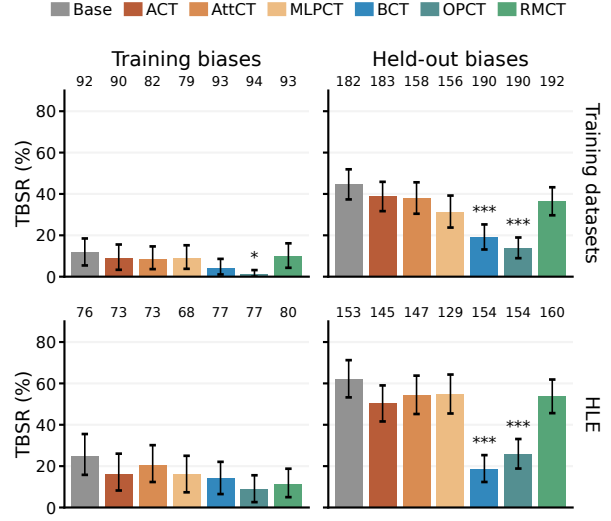


Figure 4. Towards-bias switch rate on image-rendered questions for Gemma-4-12B-IT. Lower is better. Rows as in Figure 2; columns: the suggested-answer cue, a training bias, and the sampled-shape and spurious few-shot squares cues, both held out, pooled. Pairs are eligible when both responses end with a parsed answer and the unbiased answer does not match the bias. Numbers above bars are the numbers of eligible pairs. Error bars are 95% bootstrap confidence intervals over questions; significance markers denote paired permutation tests against the base model, Holm-corrected within the figure.

significantly reduce TBSR in all four conditions and RMCT in three (Appendix C).

5.3. Bias verbalisation and monitorability

Figure 5 reports BVR on the towards-bias-switch subset on Gemma-4-12B-IT. The base model verbalises the bias in most of its towards-bias switches, and no difference between RMCT and the base model is significant. BCT significantly reduces BVR in three of the four conditions and OPCT in both held-out-bias conditions. ACT, AttCT and MLPCT significantly reduce BVR only on the held-out biases on HLE. Each method’s towards-bias-switch subset differs, and for methods that strongly reduce TBSR some subsets contain fewer than ten responses.

Figure 6 reports monitorability on Gemma-4-12B-IT, using GPT-5.6 Luna as a zero-shot monitor that sees only the chain of thought under the biased prompt. In paired bootstrap tests against the base model, Holm-corrected over the six methods, only BCT and OPCT significantly reduce the monitor’s AUROC ($p < 0.001$ and $p = 0.006$), and the AUROC on RMCT is close to that on the base model. Because the methods reduce TBSR by different amounts, we also evaluate earlier checkpoints of BCT, OPCT, ACT and MLPCT on Gemma-4-12B-IT, chosen so that their TBSR on the evaluation set brackets that of RMCT after 128 batches,

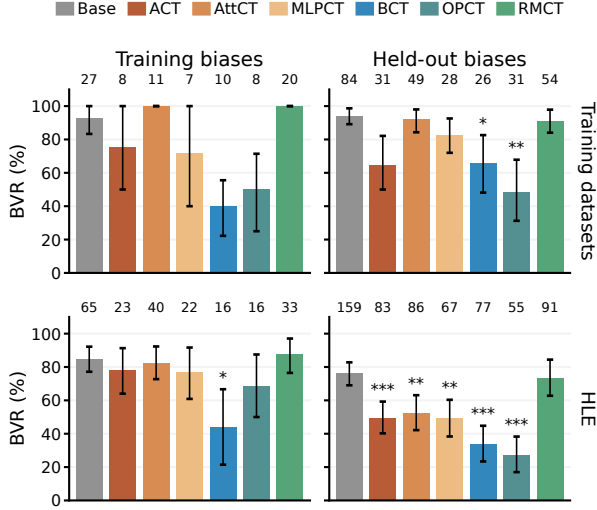


Figure 5. Bias verbalisation rate (BVR) on Gemma-4-12B-IT, restricted to the towards-bias-switch subset. Higher is better. Rows and columns as in Figure 2. Numbers above bars are the numbers of graded responses in the subset. Error bars are 95% bootstrap confidence intervals over questions. Because each method’s subset differs, significance markers denote paired bootstrap tests of the difference from the base model over questions, Holm-corrected within the figure.

and interpolate each method’s AUROC linearly between its two bracketing checkpoints (Figure 7). At RMCT’s TBSR, only BCT and OPCT have significantly lower AUROC than RMCT (0.745 and 0.721 against 0.849; Holm-corrected $p \leq 0.006$).

The monitor receives neither the question, the unbiased response nor the label. Pairs whose unbiased answer already matches the bias are included as negatives, so agreement between the answer and the bias does not by itself identify a switch. An observed switch is a single sampled outcome, and so is not ground truth for whether the bias influenced the response.

On Qwen3.5-9B, RMCT verbalises the bias in all of its towards-bias switches in every condition, and BCT and OPCT are the only methods that significantly reduce the monitor’s AUROC (Appendix C).

6. Discussion

Generalisation All six methods reduce TBSR on the held-out biases and on the held-out dataset on both models. The only non-significant reduction is RMCT’s on the training biases on the Gemma-4-12B-IT training datasets, which may indicate that RMCT is undertrained at a budget of 128 batches. Unlike Gautam et al. (2026), who report that AttCT reduces sycophancy on held-out biases more than BCT, we find that AttCT reduces TBSR on held-out biases

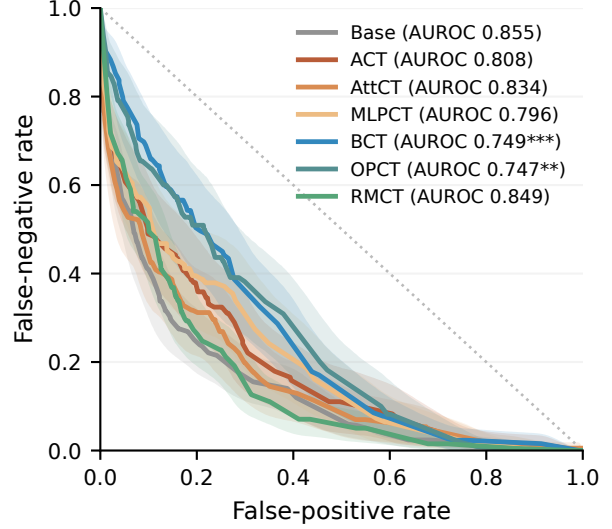


Figure 6. False-negative rate against false-positive rate for a Luna monitor on Gemma-4-12B-IT that sees only the chain of thought under the biased prompt. Lower-left is better. Positives are responses in the towards-bias-switch subset; negatives are all other responses with a parsed answer. Shaded regions are pointwise 95% bootstrap confidence intervals over questions. Stars in the legend mark AUROCs that differ significantly from the base model (paired question-cluster bootstrap, 100,000 resamples, Holm-corrected over methods; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

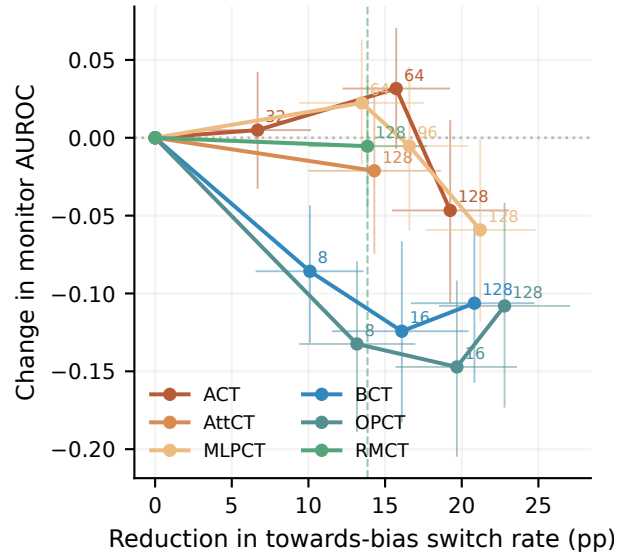


Figure 7. Change in monitor AUROC against reduction in towards-bias switch rate on Gemma-4-12B-IT, both relative to the base model, across training checkpoints. Each line runs from the base model at the origin through earlier checkpoints to the checkpoint after 128 batches. Labels give the number of training batches. The dashed line marks RMCT’s reduction. Error bars are 95% bootstrap confidence intervals over questions. Upper-right is better.

less than every other method except RMCT on Gemma-4-12B-IT, and less than BCT and OPCT on Qwen3.5-9B, although we do not test these differences between methods. Our setting differs in using reasoning models with thinking enabled, a fixed budget of 512 training questions, and the suggested-answer cue moved before the question for the activation-based methods, any of which may account for the difference.

Generalisation beyond the training format separates the methods more clearly. On AITA-NTA-FLIP on Gemma-4-12B-IT, the reduction in sycophancy carries over for the output-based methods but not for the activation-based methods: BCT, OPCT and RMCT reduce the both-NTA rate, whereas ACT, AttCT and MLPCT increase it. On Qwen3.5-9B, no method significantly changes it. Gautam et al. (2026) similarly find that transfer across threats depends on the consistency target. On image-rendered questions, BCT and OPCT reduce TBSR under the held-out visual cues on both models, whereas RMCT does so on Qwen3.5-9B but not on Gemma-4-12B-IT. On Gemma-4-12B-IT, RMCT also does not significantly reduce TBSR on the training biases on the training datasets and applies the fewest updates of any method, so its weaker transfer to images there may reflect undertraining at this budget, which we have not tested.

Monitorability On Gemma-4-12B-IT, BCT and OPCT roughly halve the proportion of towards-bias switches that verbalise the bias and significantly reduce the AUROC of a monitor reading their chains of thought. On Qwen3.5-9B, the same two methods are the only ones that significantly reduce the AUROC. This is consistent with the obfuscation concern that motivated outcomes-based consistency training: both objectives train the model to reproduce its unbiased-prompt behaviour on the biased prompt, and so penalise verbalisation of the bias. On both models, RMCT keeps the verbalisation of switches and the monitor’s AUROC close to those of the base model, consistent with its trait being defined over the final answer only, so that its reward does not depend on whether the reasoning mentions the bias. At matched TBSR on Gemma-4-12B-IT, the reduction in monitorability under BCT and OPCT persists, which indicates that it is not a consequence of reducing TBSR.

We do not yet draw conclusions about why the activation-based methods preserve monitorability at matched TBSR: their AUROC on Gemma-4-12B-IT falls between 64 and 128 batches (Figure 7), and longer training may reduce it further.

Limitations TBSR excludes responses without a parsed answer. At most 3.5% of any method’s biased responses lack a parsed answer, compared with 0.4% for the base models, and no method significantly changes clean accuracy on either model, so the reductions in TBSR are not explained

by missing answers. We report these checks, together with answer transitions and the causes of unparsed responses (Appendices F and C).

RMCT requires more training compute than the other methods at the same data budget: on Gemma-4-12B-IT, 217 GPU-hours, compared with 19 for BCT, 35 for OPCT and 0.34 for each activation-based method (Table 2). We compare all methods at a fixed budget of 512 training questions, so that every method encounters the same question–bias pairs in the same order, although batches that produce no update contribute no gradient. The methods may differ in how many training questions they need to converge, that is, to reach the point beyond which further training no longer reduces the towards-bias switch rate on held-out validation questions. Training each method until a fixed stopping rule on such validation data is met, and comparing the methods at those checkpoints, is left to future work.

Single-sample switch rates also conflate the influence of the bias with sampling variance, under which the model switches toward or away from the biased option irrespective of the cue. Repeated sampling of the unbiased response would allow estimating the switch rate expected from sampling variance alone.

Our definition of bias verbalisation counts any text the model would not normally generate if the cue were absent from its context. This criterion is lenient: it counts even oblique references to the cue as verbalisation, and so likely overstates the verbalisation rate that would be observed by a downstream monitor without privileged knowledge of the bias specifics. Our monitorability analysis, in which the monitor sees only the chain of thought, partly addresses this.

Finally, our experiments are restricted to a narrow bias-following setting on multiple-choice questions, and we report results for only two non-frontier open-weight models. Future work will examine generalisation along both axes: of the method to other inconsistency problems, such as jail-break susceptibility, persona drift, and evaluation gaming; and of the models to more capable, frontier-scale LLMs.

7. Conclusion

We introduce Rate Matching Consistency Training (RMCT), a method for outcomes-based consistency training that enforces consistency only over an outcome of interest, here whether the final answer matches the bias. On Gemma-4-12B-IT and Qwen3.5-9B, RMCT reduces the towards-bias switch rate on held-out biases and a held-out dataset while keeping the verbalisation of the bias and chain-of-thought monitorability close to those of the base model, whereas BCT and OPCT significantly reduce monitorability. Beyond the multiple-choice format, the output-based methods trans-

fer further than the activation-based methods, most clearly to a free-text judgement task on Gemma-4-12B-IT. BCT and OPCT transfer furthest, including to image-rendered questions on both models, but reduce monitorability; RMCT shares their transfer to the free-text task on Gemma-4-12B-IT while preserving monitorability, with weaker transfer to images on that model.

Acknowledgments

We thank Tim Hua for inspiring this project and Igor Ivanov for project management. We also thank Jordan Taylor, James Chua, Daniel Tan, and Jasmine Li for helpful discussions that improved this paper. We thank the London Initiative for Safe AI for hosting SI and JE and enabling this collaboration, the Supervised Program for Alignment Research for facilitating PG’s collaboration with DA and SI, and Far Labs for hosting SI. We are grateful to Coefficient Giving for funding SI and for providing compute, and to the UK AI Security Institute for providing additional compute for this research.

Author contributions

JE and SI conceptualised the method. SI implemented and ran the majority of the experiments. PG implemented the multi-turn biases and conducted hyperparameter tuning. DA supervised the project. All authors contributed to writing the paper.

References

- Africa, D. D. and Mani, A. Consistency training can entrench misalignment. In *Proceedings of the 43rd International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2026. To appear.
- Anthropic. Mitigating the risk of prompt injections in browser use. Anthropic Research, November 2025. URL <https://www.anthropic.com/research/prompt-injection-defenses>. Describes RL-based training of Claude for prompt-injection robustness.
- Baker, B., Huizinga, J., Gao, L., Dou, Z., Guan, M. Y., Madry, A., Zaremba, W., Pachocki, J., and Farhi, D. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation. *arXiv preprint arXiv:2503.11926*, 2025.
- Chen, Y., Wu, A., DePodesta, T., Yeh, C., Li, K., Marin, N. C., Patel, O., Riecke, J., Raval, S., Seow, O., Wattemberg, M., and Viégas, F. Designing a dashboard for transparency and control of conversational AI. *arXiv preprint arXiv:2406.07882*, 2024.
- Cheng, M., Yu, S., Lee, C., Khadpe, P., Ibrahim, L., and Jurafsky, D. Social sycophancy: A broader understanding of llm sycophancy, 2025.
- Chua, J., Rees, E., Batra, H., Bowman, S. R., Michael, J., Perez, E., and Turpin, M. Bias-augmented consistency training reduces biased reasoning in chain-of-thought, 2024. URL <https://arxiv.org/abs/2403.05518>.
- Drori, J., Marks, L., Woodworth, B., Cloud, A., and Turner, A. M. Output supervision can obfuscate the chain of thought, 2025. URL <https://arxiv.org/abs/2511.11584>.
- Gautam, S., Shah, N., Dhoot, A., Maruyama, B., Wei, C., Kapoor, R., Sidey, R., Gupta, P., Huang, Z. C., and Africa, D. D. Consistency training along the transformer stack, 2026.
- Han, A., Fujimoto, K., Shah, A., Nguyen, K., Xu, K., Yueh-Han, C., Sucholutsky, I., and Angell, R. On-policy consistency training improves LLM safety with minimal capability degradation, 2026.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- Hua, T. T., Qin, A., Marks, S., and Nanda, N. Steering evaluation-aware language models to act like they are deployed, 2025. URL <https://arxiv.org/abs/2510.20487>.
- Irpan, A., Turner, A. M., Kurzeja, M., Elson, D. K., and Shah, R. Consistency training helps stop sycophancy and jailbreaks, 2025. URL <https://arxiv.org/abs/2510.27062>.
- Korbak, T., Balesni, M., Barnes, E., Bengio, Y., Benton, J., Bloom, J., Chen, M., Cooney, A., Dafoe, A., Dragan, A., et al. Chain-of-thought monitorability: A new and fragile opportunity for AI safety. *arXiv preprint arXiv:2507.11473*, 2025.
- Liu, J., Cui, L., Liu, H., Huang, D., Wang, Y., and Zhang, Y. LogiQA: A challenge dataset for machine reading comprehension with logical reasoning. *arXiv preprint arXiv:2007.08124*, 2020.
- Needham, J., Edkins, G., Pimpale, G., Bartsch, H., and Hobbhahn, M. Large Language Models Often Know When They Are Being Evaluated, 2025. URL <https://arxiv.org/abs/2505.23836>.

- Nguyen, J., Hoang, K., Attubato, C. L., and Hofstätter, F. Probing and Steering Evaluation Awareness of Language Models, 2025. URL <https://arxiv.org/abs/2507.01786>.
- Perez, E., Ringer, S., Lukosiute, K., Nguyen, K., Chen, E., Heiner, S., Pettit, C., Olsson, C., Kundu, S., Kadavath, S., et al. Discovering language model behaviors with model-written evaluations. In *Findings of the association for computational linguistics: ACL 2023*, pp. 13387–13434, 2023.
- Phan, L., Gatti, A., Han, Z., Li, N., Hu, J., Zhang, H., Zhang, C. B. C., Shaaban, M., Ling, J., Shi, S., et al. Humanity’s Last Exam. *arXiv preprint arXiv:2501.14249*, 2025.
- Read, T., Schoen, B., and Bloom, J. Reproducing steering against evaluation awareness in a large open-weight model. LessWrong, April 2026. URL <https://www.lesswrong.com/posts/HhF5kESdtPHku7kim/reproducing-steering-against-evaluation-awareness-in-a-large-1>.
- Schoen, B., Nitishinskaya, E., Balesni, M., Højmark, A., Hofstätter, F., Scheurer, J., Meinke, A., Wolfe, J., van der Weij, T., Lloyd, A., Goldowsky-Dill, N., Fan, A., Matveiakina, A., Shah, R., Williams, M., Glaese, A., Barak, B., Zaremba, W., and Hobbhahn, M. Stress Testing Deliberative Alignment for Anti-Scheming Training, 2025. URL <https://arxiv.org/abs/2509.15541>.
- Shah, A., Brinkmann, J., and Angell, R. Mitigating adaptive attacks against reasoning models with activation consistency training, 2026.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y. K., Wu, Y., and Guo, D. DeepSeek-Math: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Tan, D., Chanin, D., Lynch, A., Kanoulas, D., Paige, B., Garriga-Alonso, A., and Kirk, R. Analyzing the generalization and reliability of steering vectors, 2024. URL <https://arxiv.org/abs/2407.12404>.
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4791–4800, 2019.

A. Training and evaluation details

A.1. Training data

All methods draw training questions from one pool of 7,680 questions, 3,840 from LogiQA and 3,840 from HellaSwag, traversed once in a fixed interleaved order that is shared across methods and models. Each question is assigned one of the two training biases, wrong argument or suggested answer, by a hash of its identifier, so every method sees the same question–bias pairs. Each batch contains four questions. The data-matched checkpoints are taken after 128 batches, which cover 512 questions, counting batches that produced no update. No instruction-following data is mixed in. For ACT, AttCT and MLPCT, the suggested-answer cue is moved before the question in the training prompts, following Gautam et al. (2026); all other methods and all evaluations use the original cue position.

A.2. Training methods

All methods train a LoRA adapter (Hu et al., 2022) of rank 8 ($\alpha = 16$, no dropout) with AdamW (learning rate 10^{-4} , constant schedule, $\beta = (0.9, 0.95)$, no weight decay, gradient clipping at 1.0). Thinking is enabled during training, and all sampled training responses are capped at 20,480 tokens. Table 1 summarises the method-specific settings.

For RMCT, the trait T indicates whether the parsed final answer equals the option promoted by the bias. Rollouts that reach the token limit or lack a parsed answer are excluded. On Gemma-4-12B-IT, OPCT excludes any group in which a rollout reaches the token limit; on Qwen3.5-9B, it trains on such rollouts. The number of optimiser updates within the 128 batches is therefore 110 for RMCT, 120 for OPCT and 127 for BCT on Gemma-4-12B-IT, and 118 for RMCT on Qwen3.5-9B; every other method applies 128 updates.

On Qwen3.5-9B, the ACT, AttCT and MLPCT adapters are restricted to the query and value projections of the full-attention layers.

A.3. Evaluation

The evaluation set contains 50 LogiQA and 50 HellaSwag questions, selected by a hash of their identifiers, and all 100 HLE multiple-choice questions, each under the unbiased prompt and the six biases. We sample one response per prompt with thinking enabled, temperature 1.0, top- p 0.95 and top- k 20, using a fixed seed per prompt, with vLLM. Responses are capped at 20,480 tokens on LogiQA and HellaSwag and at 65,536 tokens on HLE. The final answer is the last answer letter in the text after the reasoning. A response has no parsed answer if it reaches the token limit, does not close its reasoning, or contains no answer letter. Confidence intervals are 95% bootstrap intervals over ques-

Table 1. Method-specific training settings. The reference model is the frozen base model with the adapter disabled.

Method	Objective	LoRA modules	Sampling during training
ACT	Mean squared error between residual-stream activations on the biased prompt and those of the reference model on the unbiased prompt, over all layers and the shared prompt suffix.	Attention query and value projections	None
AttCT	Jensen–Shannon divergence between attention distributions, all layers, uniform layer weights.	As ACT	None
MLPCT	Cosine distance between SwiGLU MLP hidden activations, all layers, uniform layer weights.	As ACT	None
BCT	Negative log-likelihood, given the biased prompt, of one response sampled from the reference model on the unbiased prompt.	All attention and MLP projections	One target per question, temperature 1.0
OPCT	Per-token reverse KL divergence between the current model on the biased prompt and the reference model on the unbiased prompt, evaluated on the current model’s own samples (KL coefficient 2.0, discount factor 0.9).	All attention and MLP projections	Four rollouts per prompt, temperature 0.7
RMCT	Consistency reward (Equation 8) with the unbiased prompt as reference, optimised with a clipped policy-gradient objective (clip 0.2) and group-normalised advantages; KL coefficient 0.05; no anchor reward.	All attention and MLP projections	96 rollouts per prompt on each of the unbiased and biased prompts, temperature 1.0

tions with 10,000 resamples, stratified by dataset, and significance tests are paired permutation tests against the base model over whole questions with 100,000 permutations, Holm-corrected within each figure.

For the image evaluation, each question is rendered as an image under the unbiased prompt and three visual cues: a suggested answer, a sampled shape marking an option, and spurious few-shot squares. Sampling settings are as above.

Bias verbalisation is graded by GPT-5.6 Luna with low reasoning effort, given the bias description and the response. The monitor in Section 5 is GPT-5.6 Luna with extra-high reasoning effort, given only the reasoning of the response under the biased prompt, and asked for the probability that a cue in the prompt pulled the answer towards an option the model would otherwise not have selected.

A.4. Hardware

All training and evaluation ran on NVIDIA GH200 GPUs. ACT, AttCT and MLPCT train on one GPU. On Gemma-4-12B-IT, BCT, OPCT and RMCT use four GPUs, one for training and three for sampling. On Qwen3.5-9B, BCT and OPCT use one GPU and RMCT uses four.

B. Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see Chua et al. (2024)

for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., “My observations typically align with B.”). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An “irrelevant fun fact” about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Wrong argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as “someone else’s argument I read”, and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., “The best answer is: (B)”); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

Spurious few-shot squares. Several multiple choice questions are prepended to the test question. The correct option is always marked with a small black-square symbol, except for the test question, where an incorrect option is marked

with the same symbol. A model that picks up the extraneous symbol-to-label association will incorrectly pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistent incorrect labelling.

Excluded biases. We exclude the “Are you sure?” and positional biases, which are scored by a procedure that differs from the per-question switch rate used here and so would preclude a uniform interpretation of TBSR and BVR across bias types, and hindsight, which is supported only on a single source dataset that we did not adopt in favour of the harder HLE benchmark.

C. Qwen3.5-9B results

We repeat the comparison on Qwen3.5-9B with the same training data, budget of 128 batches, evaluation panel and statistics as on Gemma-4-12B-IT.

Figure 8 reports the towards-bias switch rate. All six methods significantly reduce TBSR in all four conditions, and no method significantly changes clean accuracy (Figure 23b).

Among its towards-bias switches, RMCT verbalises the bias in every condition, whereas BCT and OPCT significantly reduce verbalisation on the held-out biases on HLE (Figure 11). Over all biased responses, every method except RMCT significantly reduces BVR in most conditions (Figure 17b). Only BCT and OPCT significantly reduce the monitor’s AUROC (Figure 12; paired bootstrap, Holm-corrected over six methods, $p < 0.001$ for both). Per percentage point of TBSR reduction, BCT and OPCT reduce the AUROC significantly more than RMCT ($p < 0.001$ for both; Figure 13).

Figure 10 reports the image-rendered results. BCT and OPCT significantly reduce TBSR in all four conditions, and RMCT in all but the suggested-answer cue on HLE. ACT, AttCT and MLPCT significantly reduce TBSR under the held-out cues on the training datasets. On AITA-NTA-FLIP (Figure 9), no method significantly changes the both-NTA rate. Between 146 and 283 pairs per model have no valid verdict, almost all because a response reached the 65,536-token limit, and are dropped.

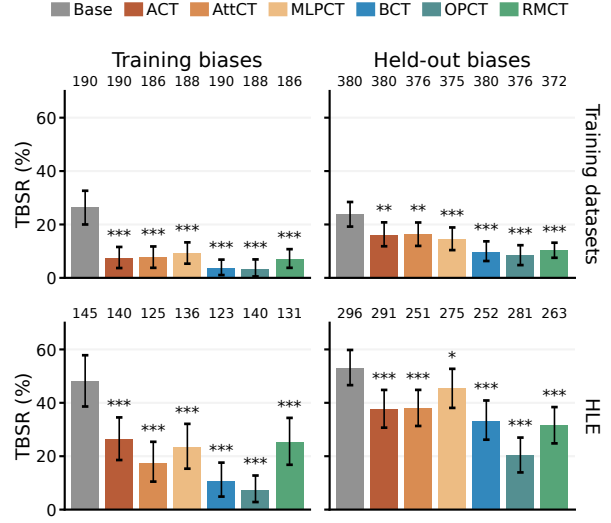


Figure 8. Towards-bias switch rate on Qwen3.5-9B. Lower is better. Rows and columns as in Figure 2. Numbers above bars are the numbers of eligible question-bias pairs. Error bars are 95% bootstrap confidence intervals over questions, stratified by dataset; significance markers denote paired permutation tests against the base model, Holm-corrected within the figure.

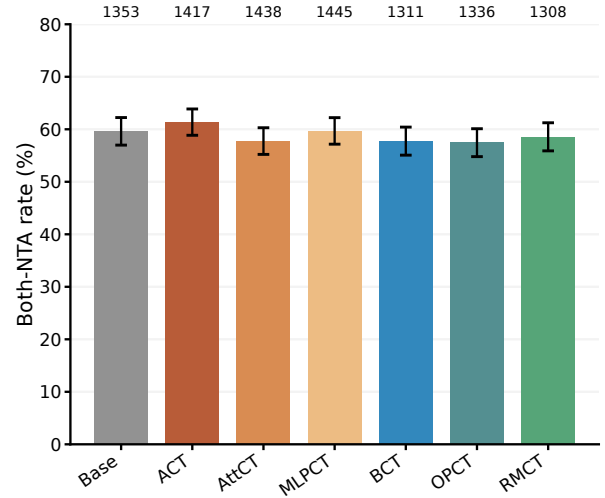


Figure 9. Both-NTA rate on AITA-NTA-FLIP for Qwen3.5-9B. Lower indicates less sycophancy. Pairs and statistics as in Figure 3.

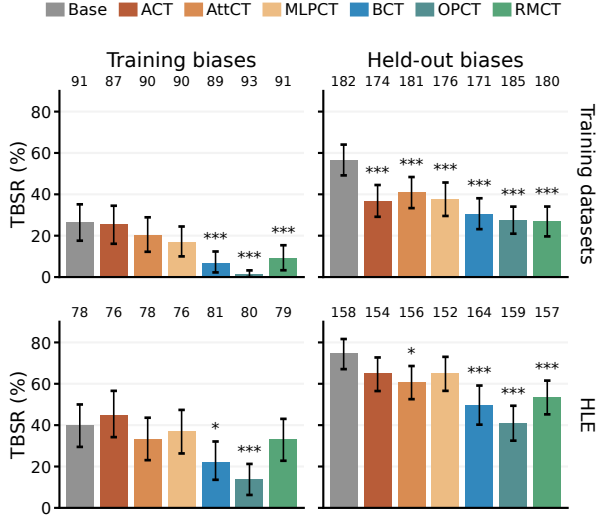


Figure 10. Towards-bias switch rate on image-rendered questions for Qwen3.5-9B. Lower is better. Columns, eligibility and statistics as in Figure 4. Error bars are 95% bootstrap confidence intervals over questions; significance markers denote paired permutation tests against the base model, Holm-corrected over the 36 comparisons in the figure.

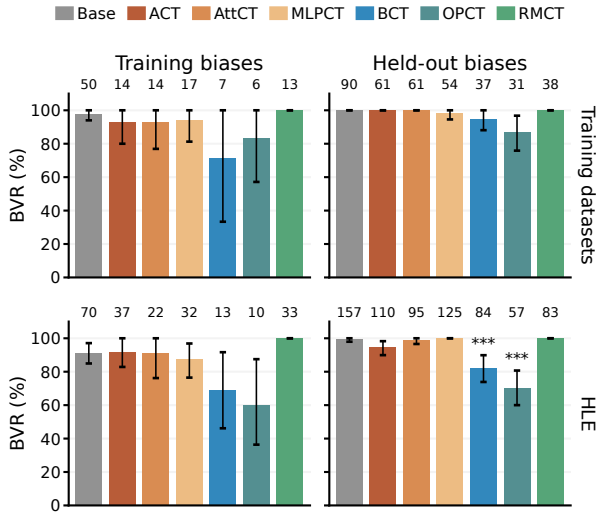


Figure 11. Bias verbalisation rate (BVR) on Qwen3.5-9B, restricted to the towards-bias-switch subset. Higher is better. Error bars are 95% bootstrap confidence intervals over questions. Because each method’s subset differs, significance markers denote paired bootstrap tests of the difference from the base model over questions, Holm-corrected within the figure.

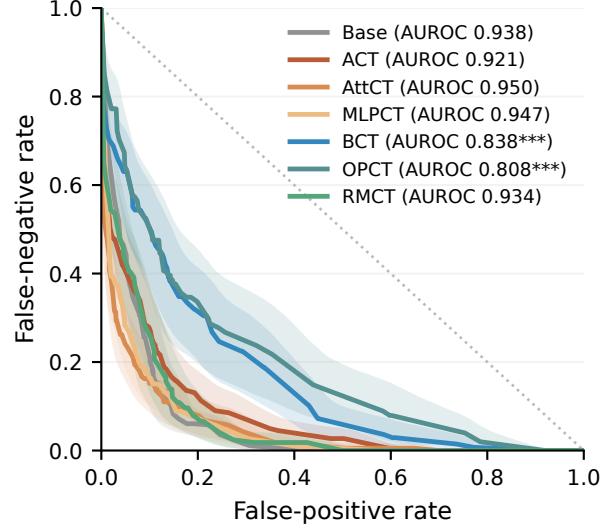


Figure 12. False-negative rate against false-positive rate for the Luna monitor on Qwen3.5-9B. Lower-left is better. Population and statistics as in Figure 6.

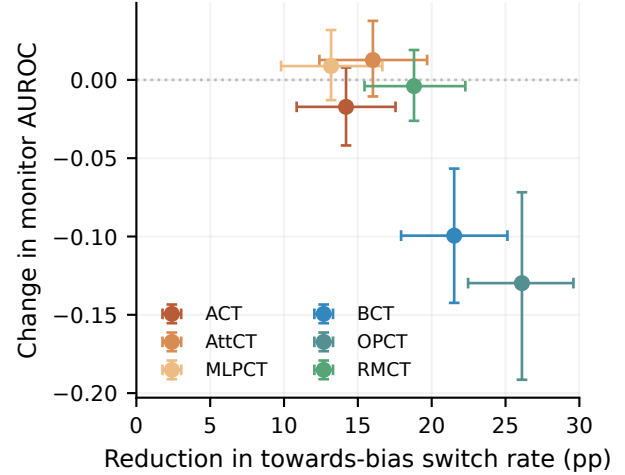
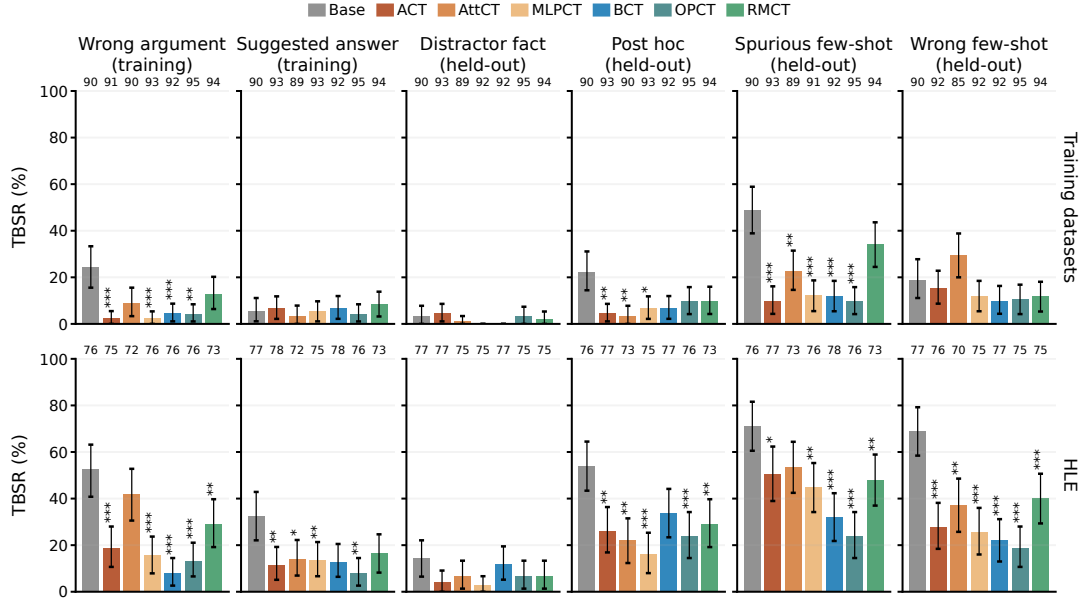


Figure 13. Change in monitor AUROC against reduction in towards-bias switch rate on Qwen3.5-9B. Upper-right is better. Statistics as in Figure 7.

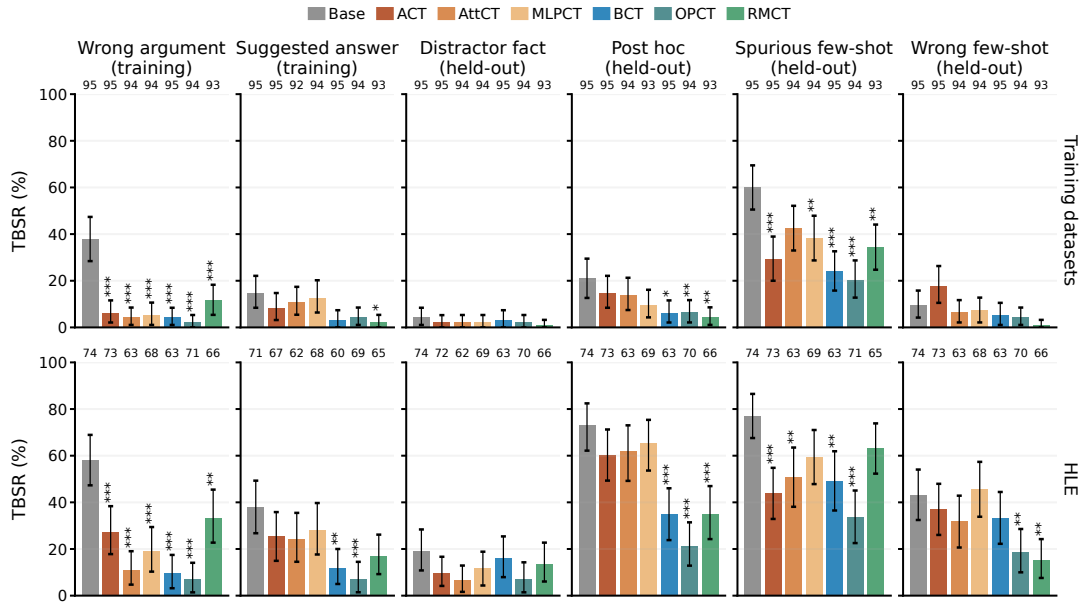
D. Per-bias switch results

Figure 14 reports the towards-bias switch rate for each bias, and Figure 15 reports it on image-rendered questions for each visual cue.

Comparison with the text cues. The suggested-answer and spurious few-shot squares cues have text counterparts evaluated on the same questions (Figure 14). The comparison below is descriptive: significance is tested against the base model within each modality, with Holm correction within each figure, and the two modalities are not tested



(a) Gemma-4-12B-IT



(b) Qwen3.5-9B

Figure 14. Towards-bias switch rate for each bias. Statistics as in Figure 2.

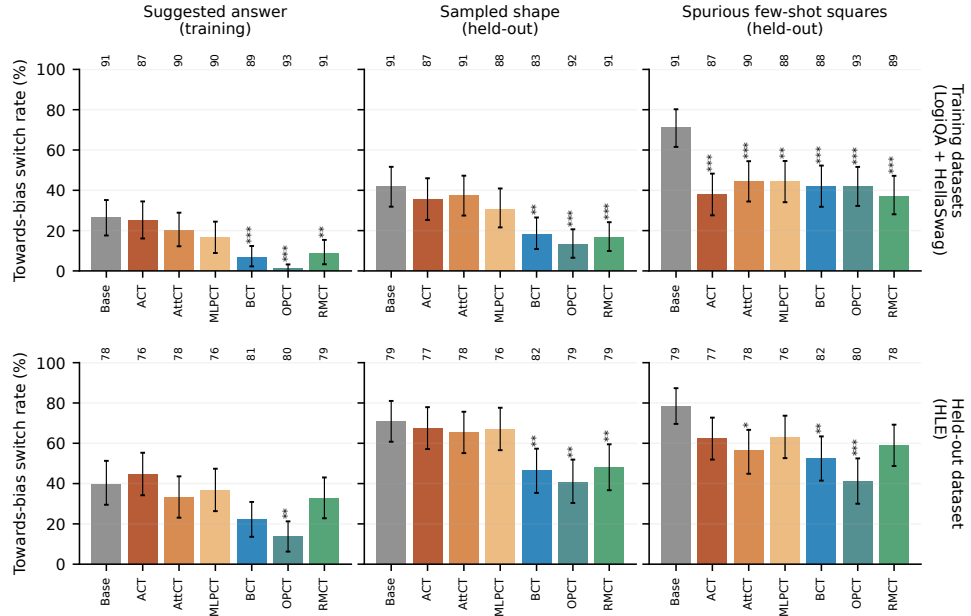
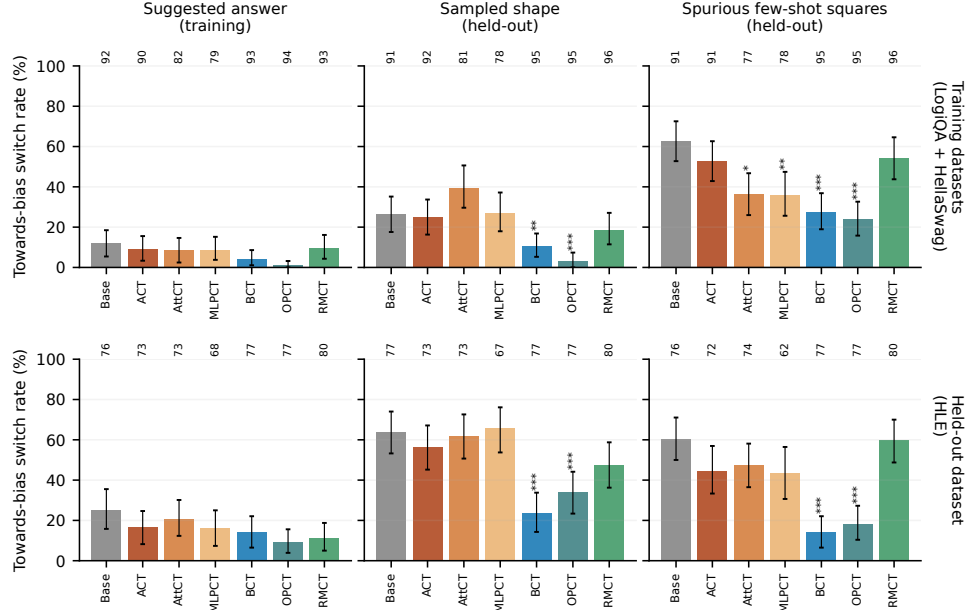


Figure 15. Towards-bias switch rate on image-rendered questions for each visual cue. Statistics as in Figure 4.

against each other. On the training datasets, the base model switches towards the bias more often under the image cues than under their text counterparts on both models, whereas on HLE the rates are similar or lower under the image cues. Reductions are generally smaller under the image cues.

On Gemma-4-12B-IT, ACT, AttCT, MLPCT and OPCT significantly reduce TBSR under the text suggested-answer cue on HLE, but no method does so under the image cue. Under the spurious few-shot squares cue, most methods significantly reduce TBSR in text on both datasets, whereas under the image cue only BCT and OPCT do so on both datasets, and AttCT and MLPCT on the training datasets only. The large text reductions for ACT on the training datasets and for RMCT on HLE do not carry over to the image cue.

On Qwen3.5-9B, under the suggested-answer cue, only RMCT significantly reduces TBSR on the training datasets in text, whereas BCT, OPCT and RMCT do so under the image cue; on HLE, BCT and OPCT do so in text and only OPCT under the image cue. Under the spurious few-shot squares cue, every method except AttCT significantly reduces TBSR in text on the training datasets, and every method does so under the image cue; on HLE, ACT, AttCT, BCT and OPCT do so in text, and AttCT, BCT and OPCT under the image cue. Across both cues and both models, BCT and OPCT are the methods whose reductions most consistently carry over from text to images.

E. Further verbalisation and monitorability results

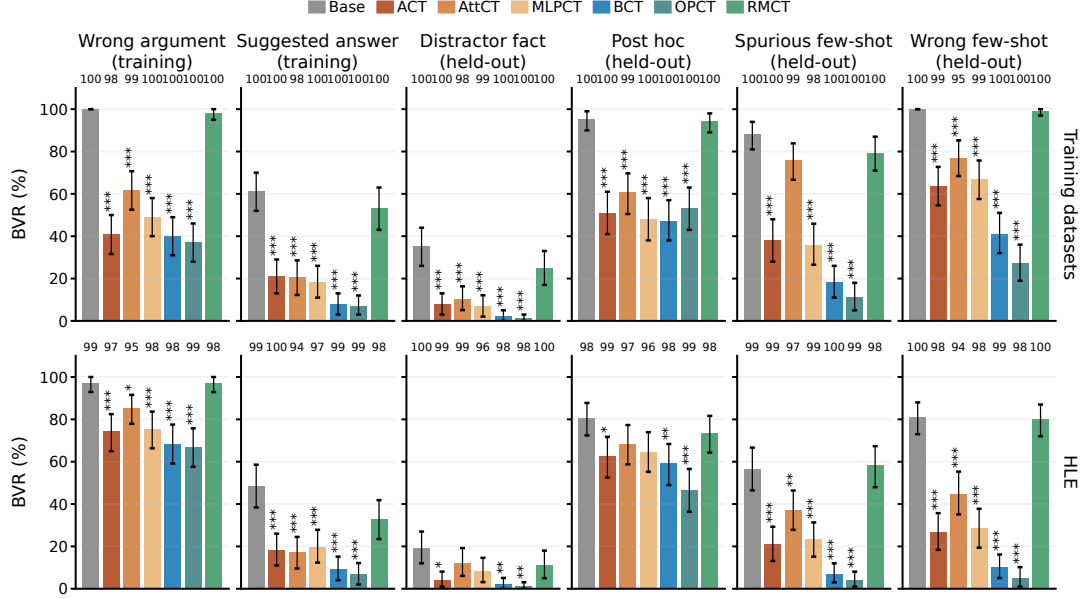
Bias verbalisation. Figure 16 reports the bias verbalisation rate for each bias. Over all biased responses, RMCT significantly reduces BVR in two of the four conditions, by at most eight percentage points, whereas every other method significantly reduces it in all four conditions (Figure 17a). Figure 18a reports, on a common population of pairs whose unbiased answer differs from the bias, the joint distribution of switching and verbalisation.

Verbalisation and monitorability. Across methods, the proportion of towards-bias switches that verbalise the bias correlates with the monitor’s AUROC: Pearson $r = 0.98$ on Gemma-4-12B-IT and $r = 0.98$ on Qwen3.5-9B, with exact permutation $p = 0.001$ and $p = 0.007$ over the seven models (Figure 19).

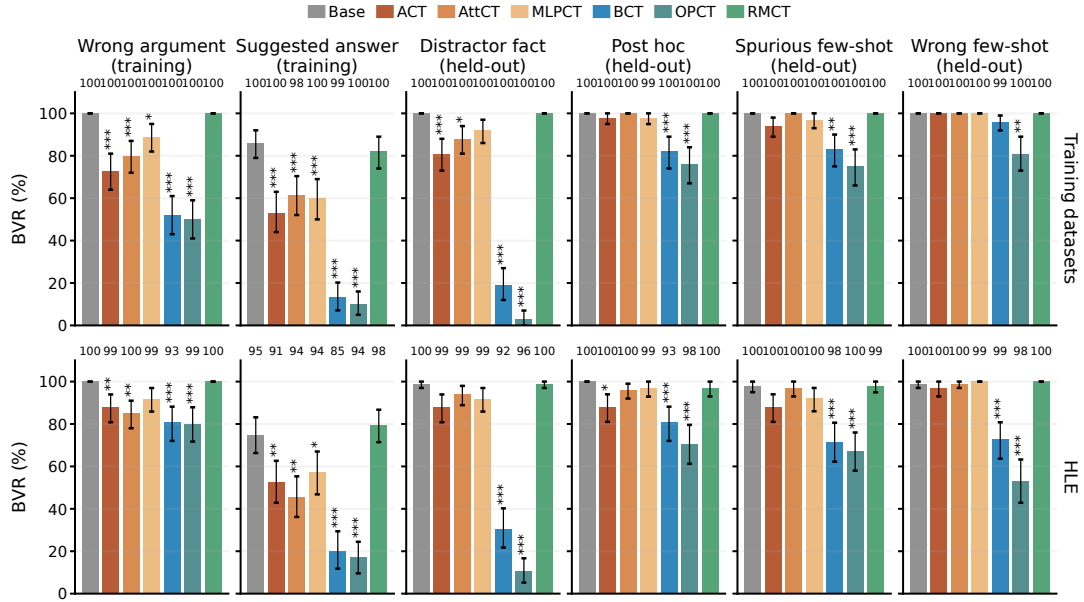
Verbalisation splits. On Gemma-4-12B-IT, on responses that verbalise the bias, the monitor’s AUROC on RMCT is close to that on the base model, whereas MLPCT, BCT and OPCT significantly reduce it (Figure 20a). On responses that do not verbalise the bias, the AUROC is lower for every model and no method differs significantly from the base

model. On Qwen3.5-9B, on responses that verbalise the bias, only BCT and OPCT significantly reduce the AUROC (Figure 20b). Fewer than 25 towards-bias switches per model do not verbalise the bias, and none for RMCT.

Matched switch rates within conditions. Within individual dataset and bias conditions, methods sometimes reach similar TBSR, which allows a comparison of monitorability at matched TBSR without selecting items by outcome (Figure 22). On Qwen3.5-9B, RMCT and BCT reach similar TBSR on the held-out biases on both the training datasets and HLE, and the monitor’s AUROC is significantly higher on RMCT in both conditions (Holm-corrected $p = 0.006$ and $p = 0.010$). On Gemma-4-12B-IT, the closest condition is HLE with held-out biases, where the difference in AUROC between RMCT and BCT is not significant.



(a) Gemma-4-12B-IT



(b) Qwen3.5-9B

Figure 16. Bias verbalisation rate over all graded biased responses, for each bias. Higher is better. Statistics as in Figure 5.

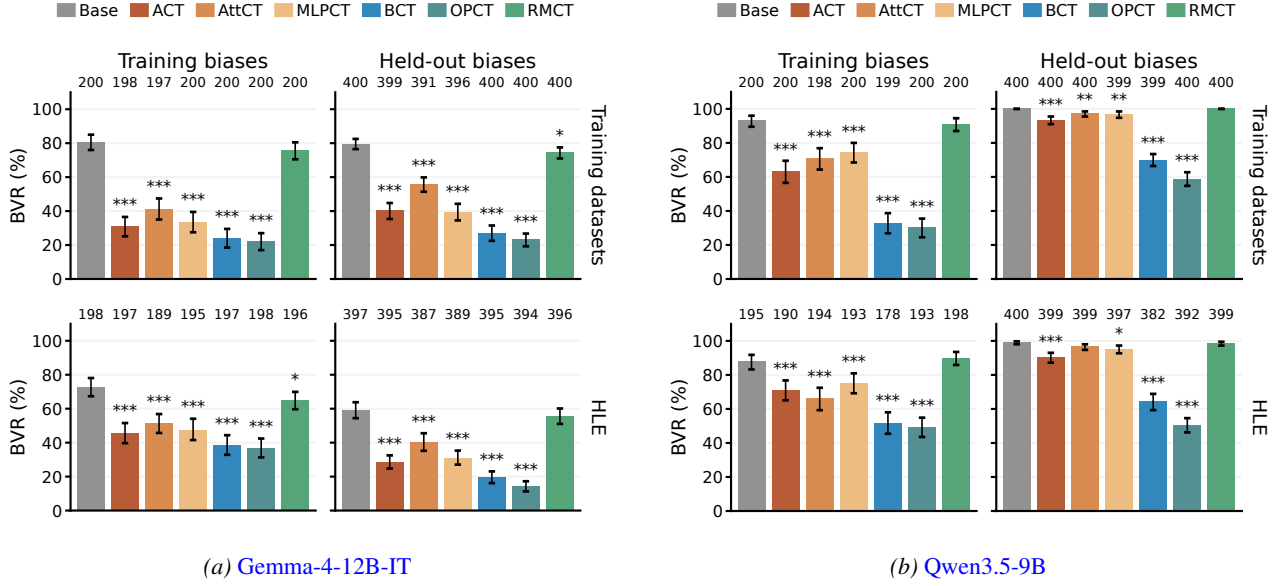


Figure 17. Bias verbalisation rate over all graded biased responses. Higher is better. Statistics as in Figure 5.

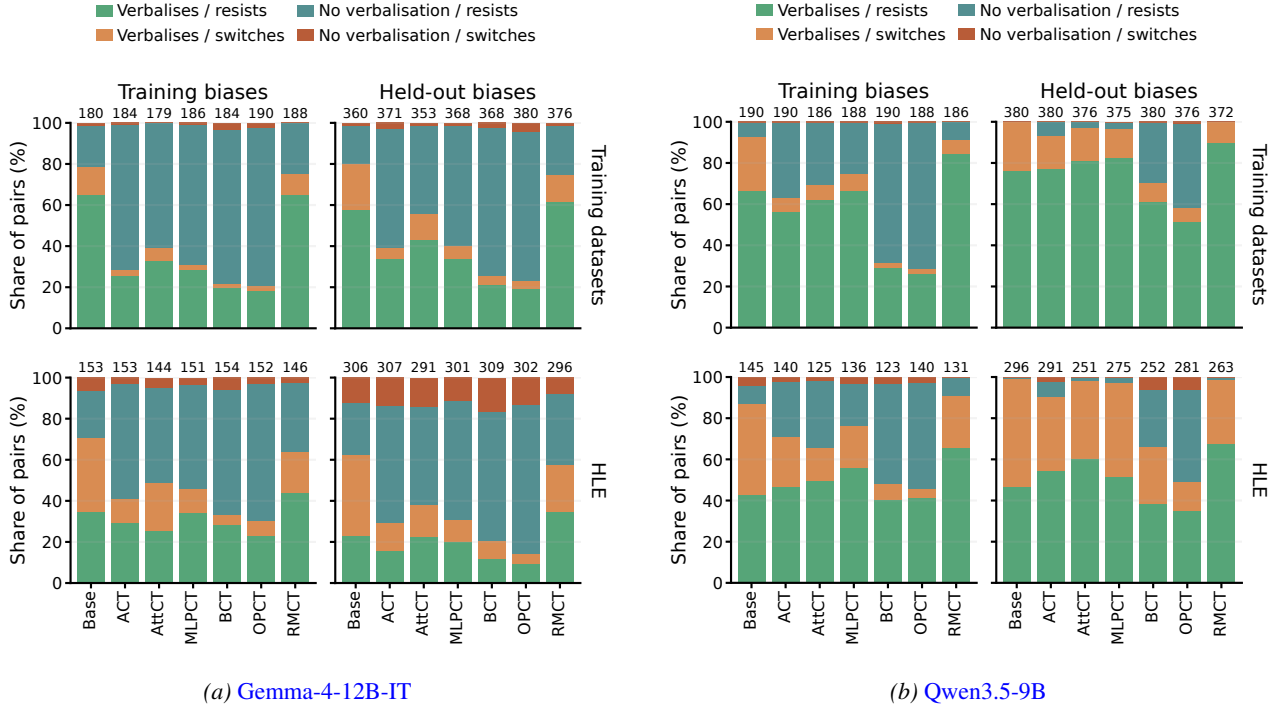


Figure 18. Joint distribution of verbalisation and towards-bias switching, over pairs whose unbiased answer differs from the bias, both answers parse, and the biased response has a verbalisation grade.

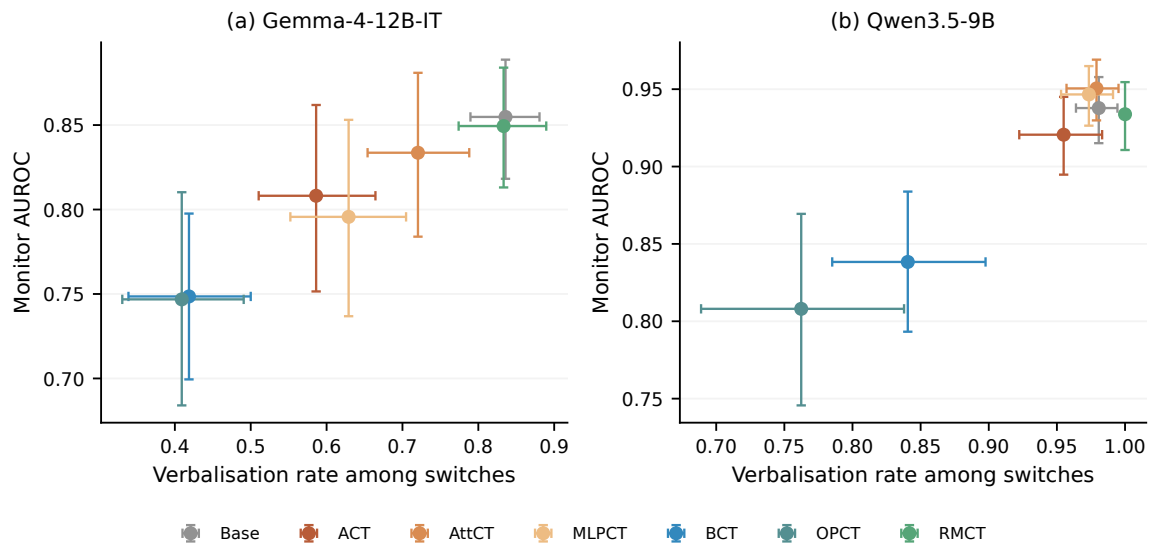
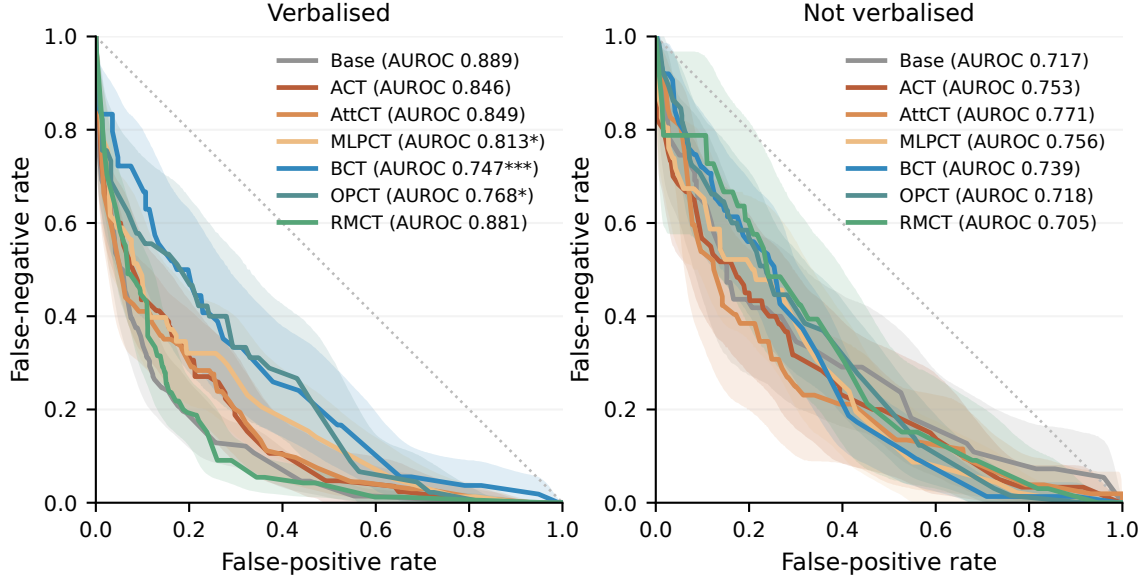
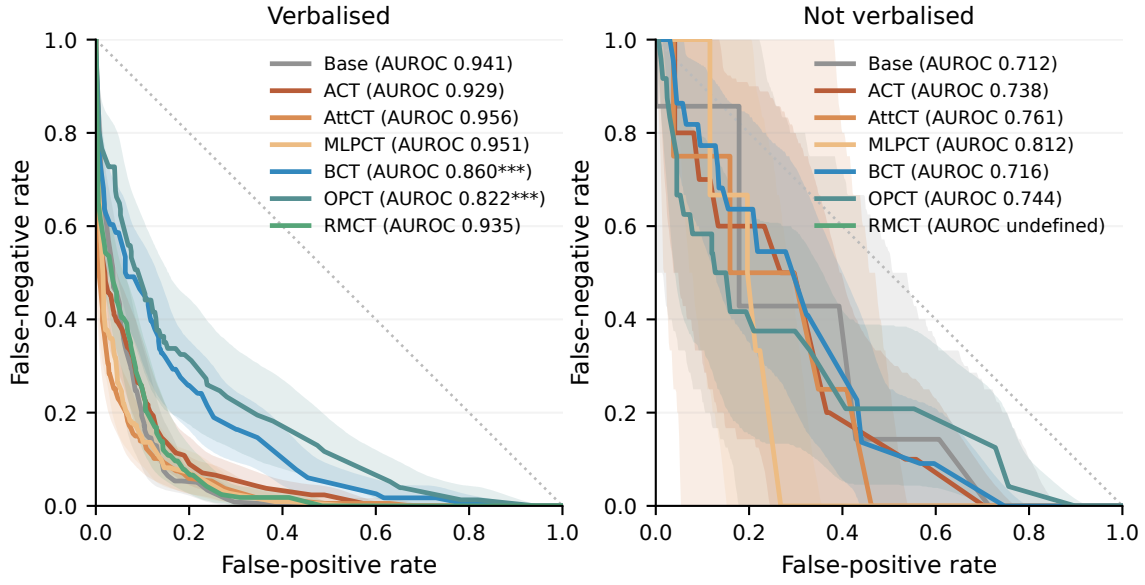


Figure 19. Monitor AUROC against the proportion of towards-bias switches that verbalise the bias, for Gemma-4-12B-IT (left) and Qwen3.5-9B (right). Error bars are 95% bootstrap confidence intervals over questions on both axes.



(a) Gemma-4-12B-IT



(b) Qwen3.5-9B

Figure 20. False-negative rate against false-positive rate for the Luna monitor, over all scored pairs, split by whether the response verbalises the bias. Lower-left is better. Significance as in Figure 6, tested within each panel.

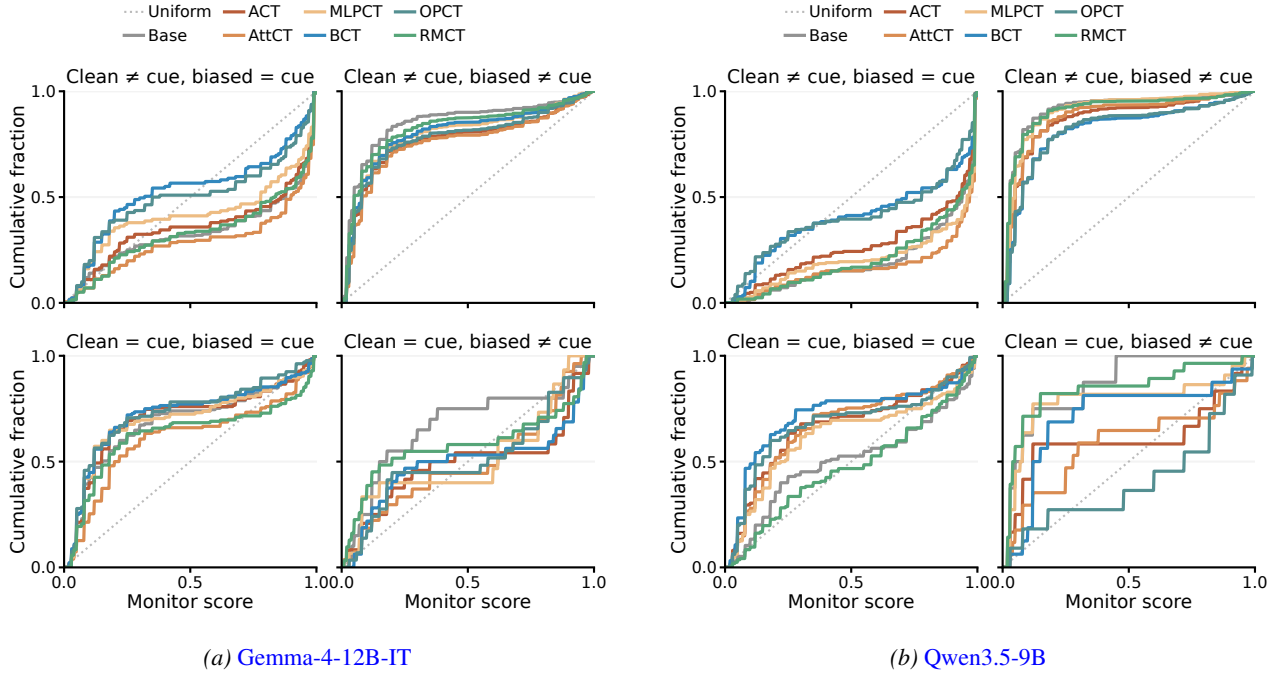


Figure 21. Luna monitor scores, shown as the cumulative fraction of responses scoring at or below each value. Each panel title states whether the unbiased (clean) and biased answers equal the cued option. The top-left panel contains the towards-bias switches, for which lower-right is better. The other three panels contain only negatives, for which upper-left is better. The dotted diagonal corresponds to uniformly random scores.

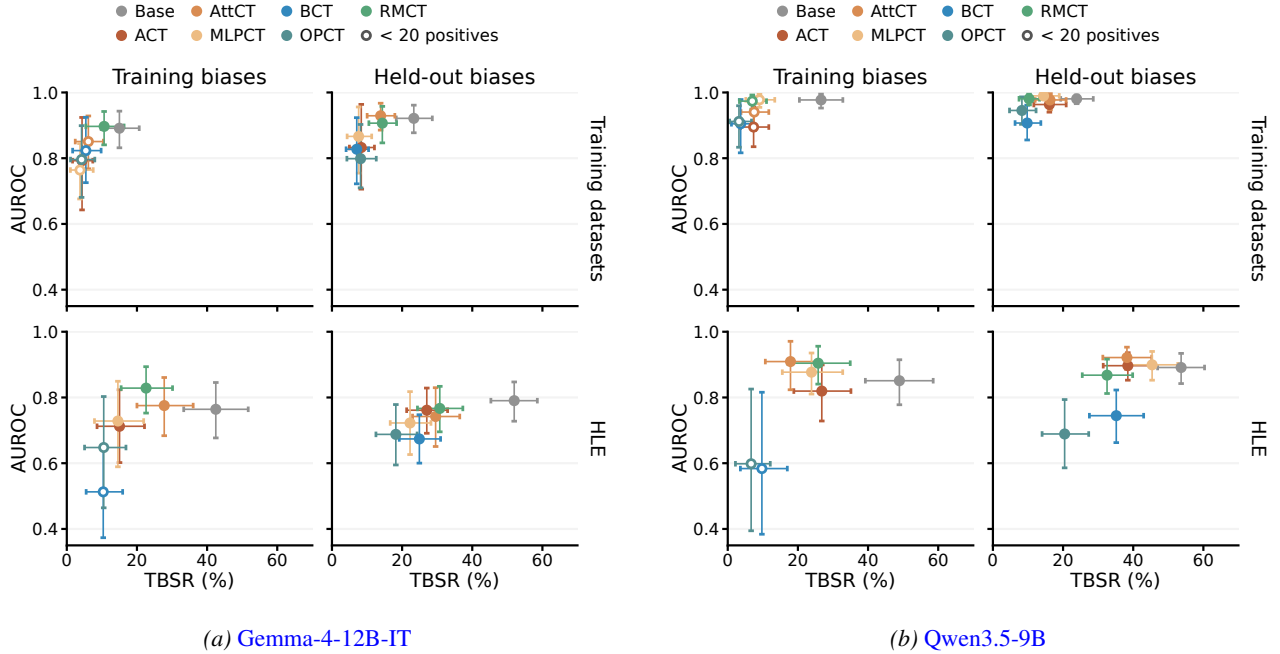


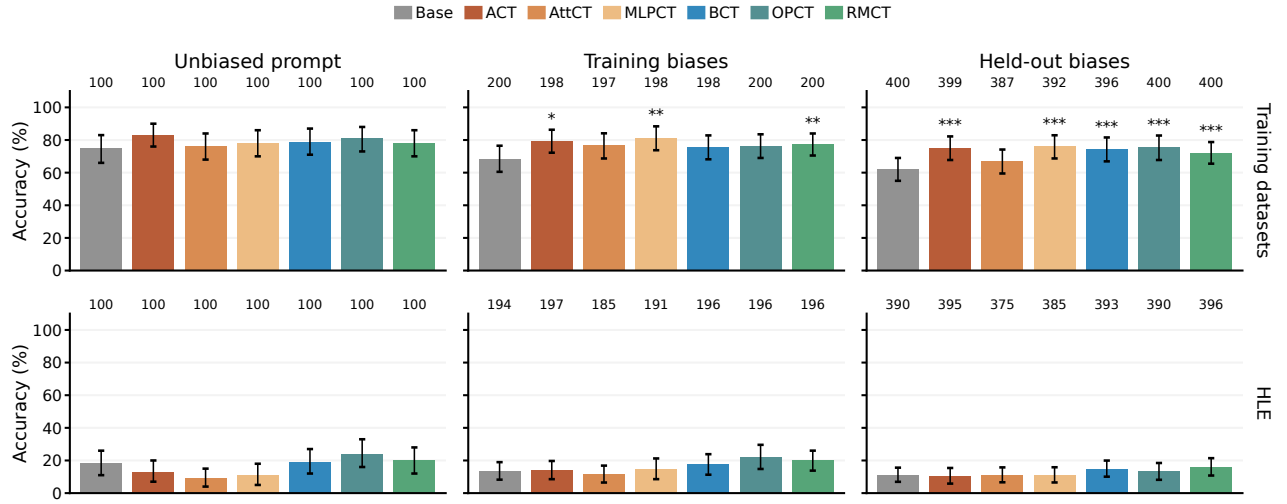
Figure 22. Monitor AUROC against towards-bias switch rate, separately for each combination of dataset group and bias group. Hollow markers have fewer than 20 positives. Error bars are 95% bootstrap confidence intervals over questions on both axes.

F. Further results

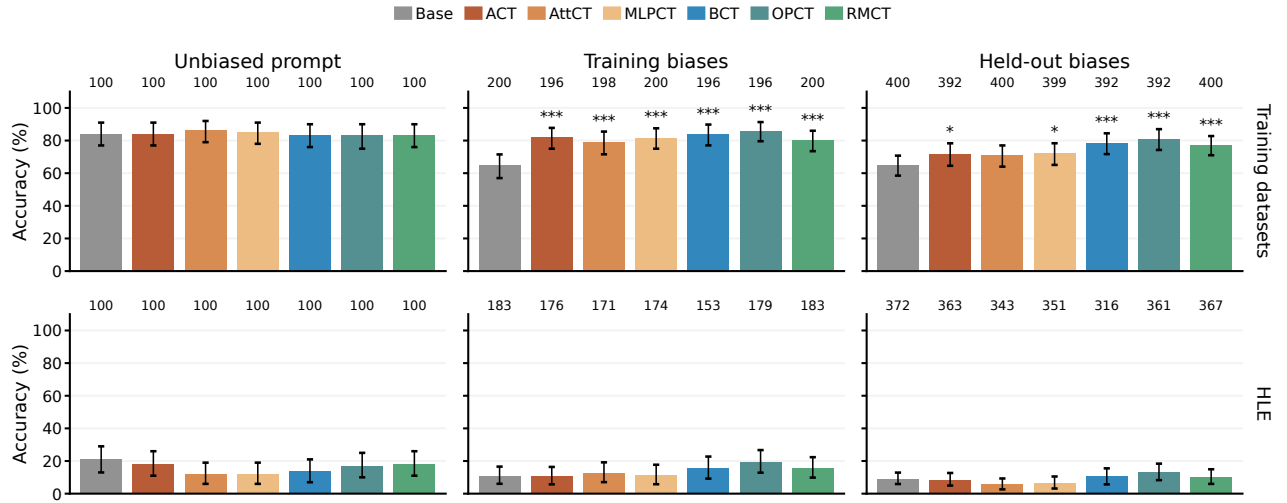
No method significantly changes clean accuracy (Figure 23a).

Among the jointly parsed pairs in which the biased answer differs from the unbiased answer, most of the base model’s changes on the training biases on the training datasets move towards the bias, whereas a smaller share do so for every trained method (Figure 24a).

Figure 27 reports the image-rendered responses without a parsed answer for both models.

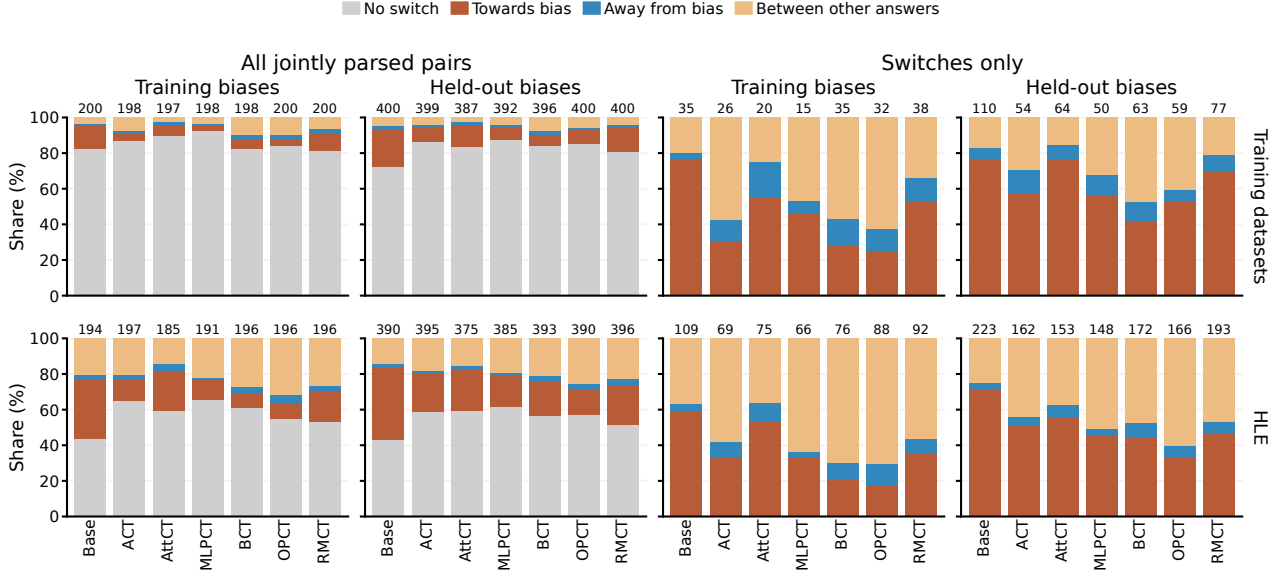


(a) Gemma-4-12B-IT

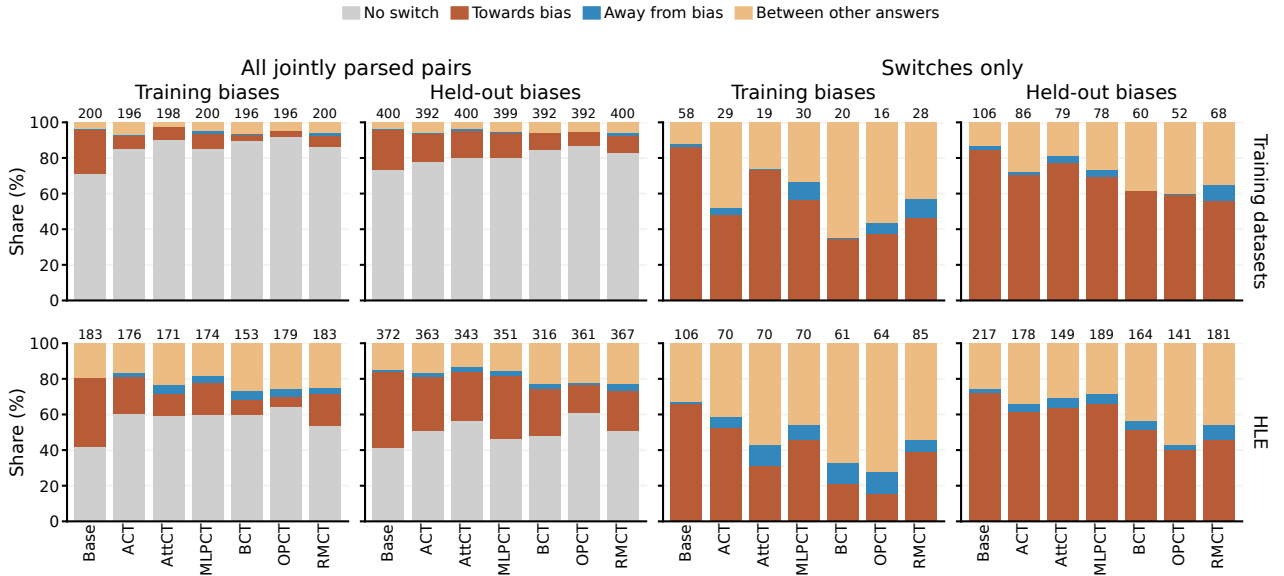


(b) Qwen3.5-9B

Figure 23. Accuracy under the unbiased prompt, counting each question once, and under the biased prompt, over all generated responses. Responses without a parsed answer count as incorrect. Statistics as in Figure 2.



(a) Gemma-4-12B-IT



(b) Qwen3.5-9B

Figure 24. Answer changes. Left: composition of jointly parsed pairs into no switch, switch towards the bias, switch away from the bias, and switch between two other answers. Right: composition of switches, over pairs in which the biased answer differs from the unbiased answer.

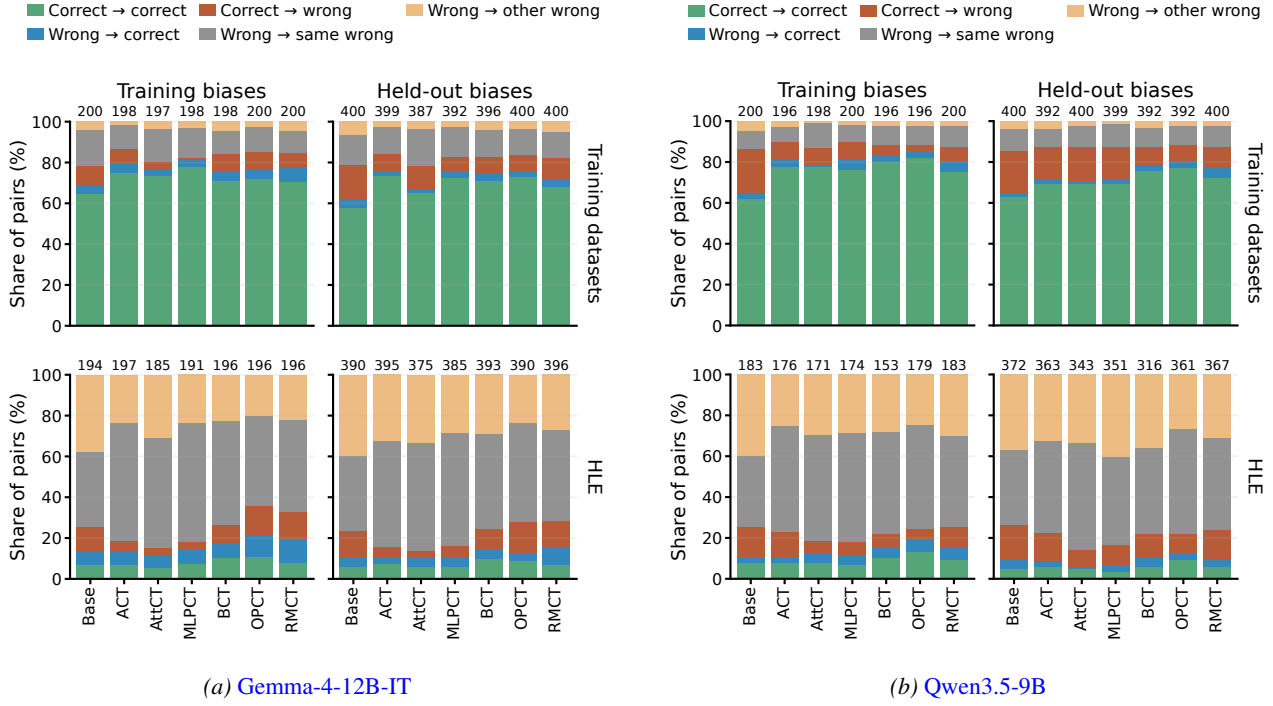


Figure 25. Transitions from the answer under the unbiased prompt to the answer under the biased prompt, over pairs in which both responses contain a parsed answer.

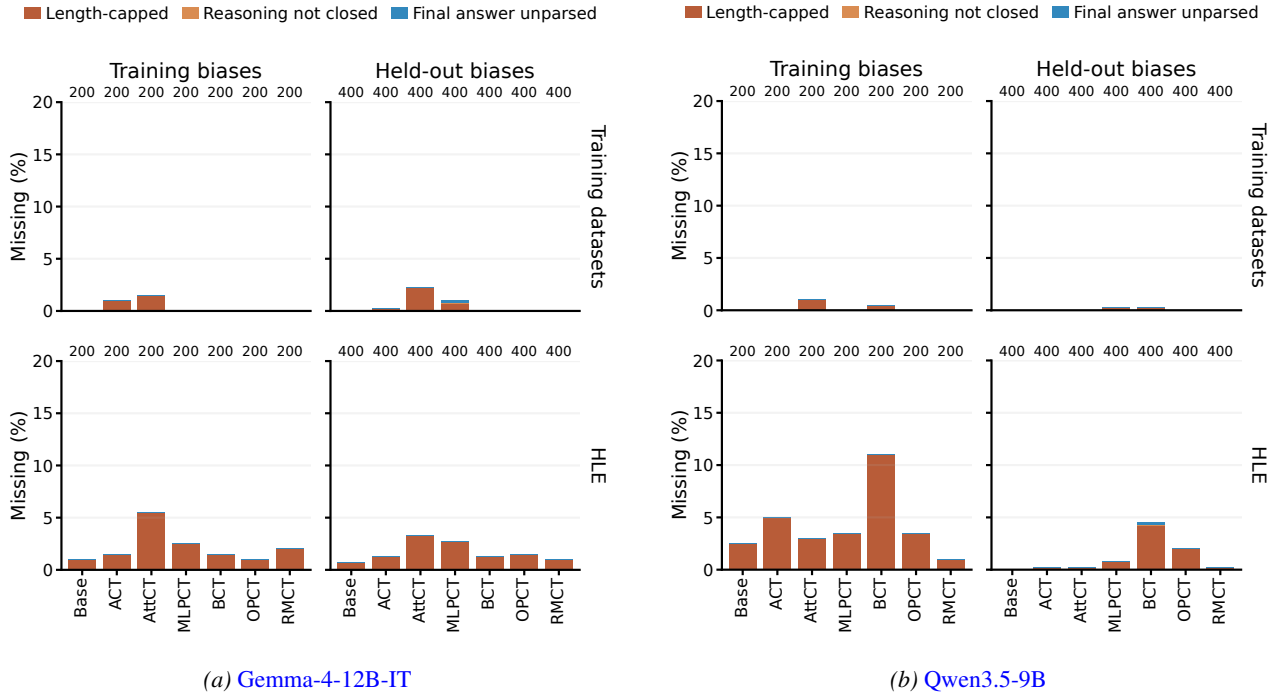


Figure 26. Proportion of biased responses without a parsed answer, by cause.

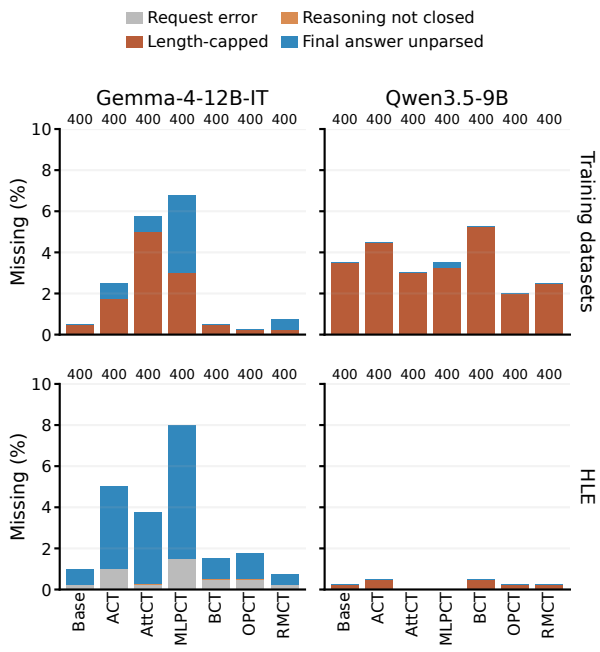


Figure 27. Proportion of image-rendered responses without a parsed answer, by cause, for Gemma-4-12B-IT (left) and Qwen3.5-9B (right), pooled over the unbiased prompt and the three cues.

G. Compute and convergence

Every method in the data-matched comparison is trained for the same 128 batches. Table 2 reports the training compute of the Gemma-4-12B-IT data-matched checkpoints. Allocated GPU-hours sum, over the training jobs that ran batches 1–128, the elapsed time multiplied by the number of allocated GPUs, and so include model loading and the generation of targets or rollouts during training. They therefore reflect each method’s hardware configuration as well as the method itself. The configurations differ between methods and between models: BCT and OPCT use four GPUs on Gemma-4-12B-IT, one for training and three for sampling, but a single GPU on Qwen3.5-9B. Ratios of allocated GPU-hours are therefore only an approximate guide to relative compute efficiency, and should not be compared across the two models. RMCT requires 643 times the allocated GPU-hours of the cheapest methods and 11.3 times those of BCT.

Table 3 reports the same quantities for the Qwen3.5-9B data-matched checkpoints. RMCT requires 464 times the allocated GPU-hours of the cheapest method, AttCT, and 3.9 times those of BCT. The activation-based methods each require less than 0.4 GPU-hours.

Table 2. Training compute for the Gemma-4-12B-IT data-matched checkpoints, each trained for 128 batches on GH200 GPUs. Allocated GPU-hours exclude bias-prompt construction, validation, evaluation and grading. Generated tokens count the sampled targets or rollouts used for training.

Method	GPUs	Updates	Allocated GPU-h	Relative	Generated tokens
ACT	1	128	0.34	1.0×	—
AttCT	1	128	0.34	1.0×	—
MLPCT	1	128	0.34	1.0×	—
BCT	4	127	19.24	57.1×	1.5M
OPCT	4	120	35.43	105.2×	7.6M
RMCT	4	110	216.65	643.0×	327.5M

Table 3. Training compute for the Qwen3.5-9B data-matched checkpoints, each trained for 128 batches on GH200 GPUs. Allocated GPU-hours include model loading and generation during training, and exclude bias-prompt construction, validation, evaluation and grading. Generated tokens count the sampled targets or rollouts used for training.

Method	GPUs	Updates	Allocated GPU-h	Relative	Generated tokens
ACT	1	128	0.33	1.06×	—
AttCT	1	128	0.31	1.00×	—
MLPCT	1	128	0.34	1.10×	—
BCT	1	128	36.4	117×	2.4M
OPCT	1	128	45.9	148×	8.7M
RMCT	4	118	143.8	464×	512.1M

H. Anchor reward

To prevent the target rate p_{ref} from drifting as the policy updates, RMCT supports an optional anchor reward that ties each reference rate p_x ($x \in \mathcal{X}_{\text{ref}}$) to its initial value $p_x^{(0)}$ computed under the initial policy π_{θ_0} (a checkpoint taken before consistency training begins):

$$r_{x,i}^{\text{anchor}} = -(p_x - p_x^{(0)}) (T(x, y_i) - p_x) \quad x \in \mathcal{X}_{\text{ref}} \quad (9)$$

Anchoring each p_x individually keeps $p_{\text{ref}}^{(0)} = Q(\{p_x^{(0)}\}_{x \in \mathcal{X}_{\text{ref}}})$ stable under any choice of aggregator Q .

Reference and non-reference trajectories receive different reward terms, combined by a mixing weight α :

$$r_{x,i} = \begin{cases} (1 - \alpha) r_{x,i}^{\text{cons}} & x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \\ \alpha r_{x,i}^{\text{anchor}} & x \in \mathcal{X}_{\text{ref}} \end{cases} \quad (10)$$

All experiments in this paper use $\alpha = 0$ (the consistency reward alone). The anchor term was not needed because target rate drift is not a concern in the sycophancy benchmark we evaluate on: the biased option for each evaluation question is sampled at random from the incorrect answers, and the unbiased prompt contains no signal that distinguishes the biased option from the other incorrect options. The target rate p_{ref} can therefore only drift via an overall shift in the unbiased-prompt answer distribution, for instance through a change in accuracy. **No method significantly changes accu-**

racy under the unbiased prompt on either model (Figure 23), so we do not expect p_{ref} to drift appreciably.

I. Extension to distribution matching

When $T(x, y) \in \{0, 1\}$, the distribution of T under any policy is a Bernoulli parameterised entirely by the rate p_x , so matching rates across inputs is equivalent to matching distributions. For a non-binary T — whether taking values in a finite set $\mathcal{T} = \{t_1, \dots, t_K\}$ (e.g. a graded quality score) or in a continuous range (e.g. a scalar reward signal) — the marginal distribution has more degrees of freedom than its mean, and matching means no longer implies matching distributions.

The framework extends naturally to finite-valued T . Define a binary indicator for each value:

$$T_k(x, y) = \mathbf{1}[T(x, y) = t_k], \quad k = 1, \dots, K. \quad (11)$$

Each T_k is a valid binary trait, so the RMCT reward can be applied to each independently and summed:

$$r_{x,i}^{\text{dist}} = \sum_{k=1}^K -(p_{x,k} - p_{\text{ref},k}) (T_k(x, y_i) - p_{x,k}), \quad (12)$$

where $p_{x,k} = \frac{1}{N} \sum_i T_k(x, y_i)$ is the empirical rate for value t_k . This is the policy-gradient estimator for the objective $\frac{1}{2} \sum_k (p_{x,k} - p_{\text{ref},k})^2$, i.e. the squared ℓ_2 distance between the empirical distributions of T on x and on the reference.

When $K = 2$, the two indicators T_1 and $T_2 = 1 - T_1$ are redundant and the reward reduces to twice the binary RMCT reward.

For continuous T , the same idea applies by replacing the sum over discrete values with an integral over thresholds t : applying the RMCT reward to the binary thresholding functions $\mathbf{1}[T(x, y) \leq t]$ for all t and integrating yields the objective $\frac{1}{2} \int (\hat{F}_x(t) - \hat{F}_{\text{ref}}(t))^2 dt$ (the squared Cramér distance between the empirical distributions), which enforces matching of the full empirical CDF.